FINAL REPORT

FORMAL OUT-OF-SERVICE API 653 TANK INSPECTION
TANK # 201, PORT EVERGLADES, FL

Prepared for:

TRANSMONTAIGNE PRODUCT SERVICES, INC.
200 MANSELL COURT EAST
SUITE 600
ROSWELL, GEORGIA 30076

JUNE 2011

SECTION 1

SIGNIFICANT FINDINGS

FOUNDATION

- Water runoff is toward the tank on the west and south sides. Also, water holds around the tank on the northwest side, after a rain. Recommend monitoring and rework the grade to improve the drainage, if conditions worsen.
- The chime area is covered with insulation and grade. Also, the chime area sits approximately 6" below the grade and the tank appears to have settled.

FLOOR

- The MFE / visual exam revealed a total of three-hundred and thirty six (336) recordable indications. Breakdown as follows: (333) Topside pitting areas with a remaining plate thickness of 0.120" or less and (03) Underside corrosion areas with a remaining plate thickness of 0.150" or less. Note: The topside pitting areas are marked with a round pink paint border. The underside corrosion areas are marked with a square pink paint border. Due to the severe topside pitting, serious consideration should be given to replacing the bottom. If the bottom is not replaced, recommend repairing the areas and installing a thick film fiberglass liner.
- Note: The floor plates are single row lap riveted. The visual exam was limited, due to the epoxy coating around the rivets. However, several areas of minor thinning were noted on the rivet heads. It is not recommended to weld thinning rivets, however insure the rivet seams are sealed with epoxy, when the new liner is installed.
- Note: The floor-to-shell rivet angle (L4") is in good structural condition. No significant thinning of the angle was noted.
- Note: The random UT thickness readings and prove-up of the MFE indications revealed the floor to be 0.250". Several of the sketch plates had a thickness between 0.255" to 0.271". For inspection purposes 0.250" was used for the calculations and pit gauging.
- Several minor-to-moderate floor plate voids were noted on the sketch and floor plates. The voids have depressions of less than one inch (1") and bottom out on the foundation below. Recommend reevaluating at the next out-of-service interval.
- The two (2) sump bottoms (welded) have a nominal thickness of 0.1875" (3/16"). The sump walls are 0.375" nominal thickness. No corrosion was noted inside the sumps. Note: This is not a recommended practice to have sump bottoms with this thickness. Consideration should be given to replacing the sump bottoms with thicker plate.
- Note: No edge settlement noted.

FLOOR (Cont.)

- Note: The tank bottom has a thick epoxy coating on the lap rivet seams (6" each side). The epoxy has varying thicknesses and is broken-up or completely missing in numerous areas. Also, the floor plates are coated with a very thin film primer coat (approximately 5 mils). The primer coat is in good condition overall with failure areas noted at the seams.
- The nineteen (19) fixed roof columns sit on the floor with no bearing plates or lateral retaining clips. Recommend installing seal welded bearing plates, under the column base pads (10"x 13") and installing lateral retaining clips. Note: The outer row of columns (12 total) have three I4" beams, welded to the bottom of the base pads. This is not a recommended practice.
- The fixed roof column at floor plate #78 has a minor twist. It is believed the column can be corrected, when the bearing plate and lateral clips are installed.
- The heating coil piping supports have 5" diameter support pads, that are not seal welded (stitch welded only). It is highly recommended that the support pads (165 total) be seal welded, before installing the new floor liner. Note: Several areas of significant topside corrosion were noted extending under the pads.
- Recommend seal welding the 8" diameter support pad for the temperature probe support (floor plate #22).
- The 2" pipe support for the 8" truck suction pipe (plate #35) is welded to the floor and pipe, allowing no movement. Recommend modifying the support to allow movement of the piping and installing a seal welded support pad on the floor. Also, recommend installing a product drainage notch, in the bottom of the support.

SHELL

- The radial stairway is in good structural condition with minor coating failure. Recommend monitoring and recoat the stairway when conditions warrant. Note: The vertical ladder (on the internal and external shell) has no fall cage. Consideration should be given to installing a fall cage on the ladder, if the ladder is to be used.
- Dried product is noted on the shell plates on the west side of the tank, where the insulation was removed (3'x 3' area). It is believed, the product is weeping from the riveted shell seams. Recommend performing a visual exam, if the insulation is ever removed.
- The visual exam of the shell coating was very limited, due to the insulation. However, the coating has hairline cracking, fading and peeling at the inspection ports and areas where the insulation was cut and removed.
- Note: No significant internal corrosion or pitting noted. Also, no significant thinning was noted at the inspection ports.
- The tank is not grounded. Consideration should be given to grounding the tank.
- No ID / data placard is visible on the tank. Consideration should be given to installing a data placard on the tank. Note: The NFPA (hazard) placards are in good visible condition.

SHELL (Cont.)

- The shell insulation is in fair condition overall. The insulation is distorted / damaged, around the heating coils on the northwest side of the tank. Also, the insulation and shell were wet on the top three (3) courses (at the inspection ports). Recommend sealing the insulation angle at the top of the tank, when the roof is replaced. Note: Recommend sealing all insulation penetrations with caulking (nozzles / support angles) when all repairs are complete.
- Recommend a complete visual exam of the external shell, rivets, nozzles and repads, if the insulation is ever removed. The visual exam was very limited, at the time of the inspection.

  Note: The shell course configurations are as follows:
- 1st and 2nd courses: 4 row / butt riveted with backing plates
- 3rd course: 4 row / lap riveted
- 4th and 5th courses: 3 row / lap riveted
- 6th and 7th courses: 2 row / lap riveted
- Horizontal seams are single row lap riveted

NOZZLES AND APPURTENANCES

- Note: The two (2) waterdraw nozzles have typical gate valves.
- Inlet nozzle "X" (16") has an older design gate valve. The valve appears to be in good condition. However, consideration should be given to installing a newer design gate valve, while the tank is out-of-service.
- The valve for suction nozzle "I" has a heavy product stain at the valve stem. Recommend monitoring or repair while the tank is out-of-service.
- The pressure relief piping for suction nozzle "I" (at pop-off) has a heavy product stain at the threads. Recommend repairing the piping while the tank is out-of-service. Note: The pressure relief system for nozzle "I" has the older design crossover piping around the valve. Consideration should be given to installing a typical TransMontaigne design system, which empties back into the tank with a 2" shell nozzle.
- Inlet nozzles "J" (10") & "X" (16") and truck suction nozzle "Q" (8") have no pressure relief systems at the valves. Recommend installing typical TransMontaigne design pressure relief systems at these nozzles.
- Note: The mixer was removed from manway "A". Minor product stains were noted at the swivel. Recommend monitoring the mixer when reinstalled. Also, no independent grounding cable was noted at the manway. Consideration should be given to installing a grounding cable on the mixer, when reinstalled.
- The Varec float has been removed from the tank. Also, the Varec gauge housing is holding water and is not legible. Recommend repairing the Varec float gauge, before returning the tank to service.
- Nozzle "C" (8" / not in use) has a neck thickness (0.313"), less than recommended by API-650.
- Inlet nozzle "J" (10") has a neck thickness (0.378"), less than recommended by API-650.
- Truck suction nozzle "Q" (8") has a neck thickness (0.312"), less than recommended by API-650.

NOZZLES AND APPURTENANCES (Cont.)

- Inlet nozzle "X" (16") has a neck thickness (0.373"), less than recommended by API-650.
- Waterdraw nozzle "AA" (2") has a neck thickness (0.145"), less than recommended by API-650.
- Butt inserts "F" and "Z" (8") have dimensions, less than recommended by API-653. Note: Inserts were viewed from inside the tank.
- Low suction nozzle "H" (10") still has the 90º elbow (hard piping) in place. The low suction piping is welded to the 16" suction piping (nozzle "I") with no valve. Recommend removing the 90º low suction elbow and installing a typical TransMontaigne design removable hose.
- Temperature probe couplets "M" (2nd course), "N" (4th course), "O" (6th course) and "W" (1st course) are no longer being used. Consideration should be given to removing the couplets and internal piping, if no longer needed.
- Consideration should be given to recoating all nozzle and manway necks, flanges and covers that are not insulated. These appurtenances have coating failure with active rust discoloration.
- Suction nozzle "I" (16") was installed by the "Hot-Tap" method. The internal nozzle neck is not back welded to the shell. API-653 recommends "Hot-Taps" larger than 10" in diameter, be back welded internally. Recommend back welding before returning the tank to service.
- The 8" piping at truck suction nozzle "Q" has significant weld corrosion and coating failure at the tie-in. The area is located between the valves, 4' from the tank and is marked with a yellow paint marker. Consideration should be given to touching-up the weld in this area.
- Doorsheet "B" has square (non-rounded) corners. API-653 recommends rounded corners on butt inserts / doorsheets. Note: The doorsheet was viewed from inside the tank.
- Note: The mixer is installed in the cover of manway "A" and has no type of supports. Consideration should be given to supporting the mixer from the shell.
- Note: The inlet diffuser at nozzle "X" (16" / no flange) is 9'-3" long with slots the entire length. The slots are 16"L x 2"W (8" apart). The diffuser has a "U" bolt piping support with pads and is in good structural condition. Inlet nozzle "J" (10" / riveted) is flush with the shell and open ended. Consideration should be given to installing a diffuser at this nozzle, if needed.
- Suction nozzle "I" (16" / "hot-tap") has no type of suction trough. Consideration should be given to installing a typical TransMontaigne design suction trough, if needed. Note: The low suction piping is a 90º elbow-down design, which sits approximately 12" above the floor.

FIXED ROOF

- No exam of the roof plates or nozzles. The roof plates have severe underside thinning with hundreds of through plate holes. The thinning is through out the entire roof. Due to the condition of the roof, the decking is to be replaced at this out-of-service. Note: The floor has significant ponding after each rain.

FIXED ROOF (Cont.)

- The eave angle (L4" / Lap welded / Inward turned) has a corrosion hole, in the top of the angle. The hole is located above manway "A". Recommend repairing when the roof is replaced.
- A total of twenty (20) rafters are bent / twisted. The roof has a total of one-hundred and forty-four (144) rafters. Location of distorted rafters: (5) center section, (3) middle section and (12) outer section. Recommend replacing these rafters at a minimum. Also, recommend performing a visual exam on the rafters for thinning, when the roof decking is replaced.
- Note: The rafter and shell clip (L4" angle) above floor plate #110 was inspected by using a manlift. The rafter had no visible corrosion or thinning and the support clip was in good condition. Also, the rafters on each side appeared to be in good structural condition. However, recommend visually inspecting all rafters, girders and supports when the roof decking is replaced. It is believed the rafters could be thinning, in the areas of the through plate holes.

INTRODUCTION

Semper Fi Enterprises, LLC was contracted by TransMontaigne Product Services, Inc. to provide inspection services for Tank # 201 (North Terminal), located in Port Everglades, Florida.

This report documents the findings of the examination performed in June of 2011, as per the criteria of API-653.

Jeff McFerran Date
API Inspector #1307

TABLE OF CONTENTS

Section No.	Title	Page
1	**SIGNIFICANT FINDINGS**	**1-1**
2	**VISUAL INSPECTION AND API-653 CHECKLIST**	**2-1**
	Tank Physical and Historical Data	2-1
	Foundation Inspection	2-2
	Foundation Settlement Survey	2-3
	Bottom Inspection	2-5
	Interval To Next Bottom Inspection	2-7
	Bottom Plate Reduction Table	2-8
	Shell Inspection	2-22
	Shell Thickness Evaluation	2-24
	Shell Course Ultrasonic Thickness Readings	2-27
	Shell Nozzles and Appurtenances Inspection	2-28
	Shell Nozzles and Appurtenances Table	2-31
	Fixed Roof Inspection	2-33
3	**NON-DESTRUCTIVE TESTING (NDT)**	**3-1**
	NDT Significant Findings	3-1
	NDT Scope of Work	3-1
	Recording Criteria	3-1
	Floor Examination	3-2
	Floor Plate Lap Rivet Examination	3-2
	Floor-to-Shell Angle Examination	3-2

Shell Course Ultrasonic Thickness Readings 3-2

Shell Nozzle Examination..... 3-2

4 PICTURES 4-1

5 DRAWINGS 5-1

SECTION 2

DESCRIPTION

CUSTOMER:	TransMontaigne Product Services Inc.
TANK NUMBER:	201 (North Terminal)
LOCATION:	Port Everglades, Florida
MANUFACUTRER:	CB&I (Horton Tank)
DESIGN STANDARD:	API-12C
YEAR BUILT:	1936
PRODUCT:	#6 Fuel Oil
DIAMETER:	117'-0"
HEIGHT:	42'-0"
PRODUCT PRIOR TO INSPECTION:	#6 Fuel Oil
S.G. OF PRODUCT PRIOR TO INSPECTION:	.88
1.0 S.G. FILL HEIGHT:	36.60'
DOT REGULATED TANK:	No
FOUNDATION:	Earth
BOTTOM CONSTRUCTION:	Lap Riveted (Single Row)
SHELL CONSTRUCTION:	Butt & Lap Riveted
FIXED ROOF CONSTRUCTION:	Lap Welded
FLOATING ROOF CONSTRUCTION:	N/A
PRIMARY SEAL:	N/A
SECONDARY SEAL:	N/A
CATHODIC PROTECTION:	Yes
HEATED:	Yes (< 200° F)
NEW BOTTOM / YEAR INSTALLED:	N/A – Original Bottom
BOTTOM COATING / YEAR:	Thin Film Primer Coat
SHELL COATING / YEAR:	Painted / Green / External (Insulated)
FIXED ROOF COATING / YEAR:	Painted / Black / External
FLOATING ROOF COATING / YEAR:	N/A
STAIRWAY CONSTRUCTION:	Radial Stairway & Vertical Ladder
FLOATING ROOF ACCESS:	N/A
NOMINAL FLOOR PLATE THICKNESS:	0.250"
NOMINAL SKETCH PLATE THICKNESS:	0.250"
NOMINAL FIXED ROOF PLATE THICKNESS:	0.1875"
NOMINAL FLOATING ROOF PLATE THICKNESS:	N/A
CRITICAL RECORDING CRITERIA:	0.150" Underside / 0.120" Topside
TANK FILES AVAILABLE FOR REVIEW:	Yes
TYPE OF INSPECTION:	Formal Out-of-Service w/ MFE Floorscan
API INSPECTOR:	Jeff McFerran
DATES OF INSPECTION:	06-23-2011 / 07-01-2011

FOUNDATION INSPECTION

Item	Description	Comments
1	Foundation description	• Earth with gravel apron.
2	Inspect for foundation damage or cracking.	• Acceptable – No damage noted around the tank.
3	Evidence of bottom leakage or stains.	• Acceptable – No leaks or stains noted around the foundation.
4	List all significant cavities or erosion around the foundation.	• Acceptable – No erosion or cavities noted around the tank.
5	Inspect for vegetation under and around tank perimeter.	• Acceptable – No significant vegetation noted around the tank, in the gravel apron.
6	Does tank have proper water runoff and drainage away from the tank?	• Water runoff is toward the tank on the west and south sides. Also, water holds around the tank on the northwest side, after a rain. Recommend monitoring and rework the grade to improve the drainage, if conditions worsen.
7	Is grade or other material covering the floor-to-shell weld or chime bottom plate?	• The chime area is covered with insulation and grade. Also, the chime area sits approximately 6" below the grade and the tank appears to have settled.
8	Is any visual settlement noted around the tank perimeter?	• Acceptable – No significant visual settlement noted. Refer to the "Internal Settlement Survey" for data. Also, see Item # 7 above.
9	What is the general condition of the dike area? Berms are constructed of what material? Is dike drain area clean?	• Acceptable • Earth & gravel • Acceptable – The drain pipe and valves are in good condition.
10	Check for presence or condition of moisture barrier.	• N/A – Earth foundation and insulation.
11	Check for anchorage. Record sizes, spacing between anchors and condition.	• N/A – No anchorage system (earth foundation).

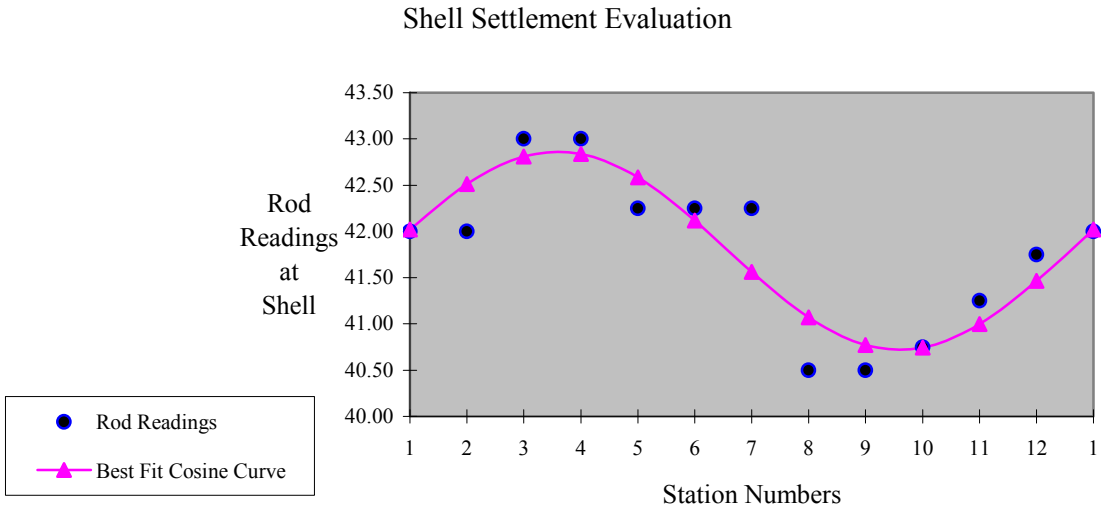
FOUNDATION SETTLEMENT SURVEY

Station #	Feet	Shell	R (ft)	R	B (in)	10'	20'	30'	40'	50'	Z	U	S
1	0	42.00	3.00	43.25	0.00	46.25	47.50	47.50	47.00	46.25	42.02	-0.02	0.24
2	30.75	42.00	3.00	43.50	0.00	46.75	46.75	47.25	47.00	45.25	42.51	-0.51	-0.60
3	61.50	43.00	3.00	43.50	0.00	46.25	46.75	47.50	47.00	45.50	42.81	0.19	0.36
4	92.25	43.00	3.00	43.00	0.00	45.25	47.50	47.00	46.50	45.75	42.84	0.16	0.23
5	123.00	42.25	3.00	42.50	0.00	45.75	46.75	46.75	47.00	45.25	42.58	-0.33	-0.48
6	153.75	42.25	3.00	44.00	0.00	46.75	47.75	46.50	46.25	45.00	42.12	0.13	-0.04
7	184.50	42.25	3.00	44.25	0.00	47.00	46.25	46.75	46.25	45.25	41.56	0.69	0.91
8	215.25	40.50	3.00	42.00	0.00	44.75	45.00	45.50	46.25	45.25	41.07	-0.57	-0.78
9	246.00	40.50	3.00	41.75	0.00	43.25	44.75	46.00	45.75	45.50	40.77	-0.27	0.01
10	276.75	40.75	3.00	42.00	0.00	45.00	45.75	45.50	45.75	45.50	40.74	0.01	0.02
11	307.50	41.25	3.00	41.75	0.00	45.75	45.25	46.50	47.50	46.25	41.00	0.25	0.11
12	338.25	41.75	3.00	43.50	0.00	46.75	46.25	46.75	47.00	46.25	41.47	0.28	0.16

Distance between stations: 30’-9”
Total Measured Circumference around tank: 369’-7”
Settlement survey stations were measured CCW from reference shell manway “A” (0’).

Z = Calculated, rigid body tilt settlement (inches)
U = Out of plane settlement (API 653, App. B) (inches)
S = Out of plane deflection (API 653, App. B) (inches)

SHELL SETTLEMENT CHART



ACCEPTABLE SHELL DEFLECTION

Yield Strength Y (psi):	30,000
Arc length betwn measured pts L (ft):	30.75
Youngs Modulus E (psi):	29,000,000
Height of Tank H (ft):	42.00

Equation (based on API 653 Section B.3.2)
allowable deflection <= (11* Y * L^2) / (2 * E * H)

allowable deflection <= 1.54

Maximum deflection = 0.91 inches < 1.54 inches; thus, actual deflection is within API-653 acceptable limits.

BOTTOM INSPECTION

Item	Description	Comments
1	Conduct visual inspection of bottom plate lap rivets.	• Note: The floor plates are single row lap riveted. The visual exam was limited, due to the epoxy coating around the rivets. However, several areas of minor thinning were noted on the rivet heads. It is not recommended to weld thinning rivets, however insure the rivet seams are sealed with epoxy, when the new liner is installed.
2	Perform vacuum box bubble test of lap welds.	• N/A – Not performed (riveted)
3	Perform visual inspection on the internal shell-to-bottom rivet angle.	• Note: The floor-to-shell rivet angle (L4") is in good structural condition. No significant thinning of the angle was noted.
4	Perform MFE floor scan and visual exam of the tank bottom.	• The MFE / visual exam revealed a total of three-hundred and thirty six (336) recordable indications. Breakdown as follows: (333) Topside pitting areas with a remaining plate thickness of 0.120" or less and (03) Underside corrosion areas with a remaining plate thickness of 0.150" or less. Note: The topside pitting areas are marked with a round pink paint border. The underside corrosion areas are marked with a square pink paint border. Due to the severe topside pitting, serious consideration should be given to replacing the bottom. If the bottom is not replaced, recommend repairing the areas and installing a thick film fiberglass liner.
5	Perform vacuum box and or a MT test on the internal shell-to-bottom weld.	• N/A – Not performed (riveted)
6	Perform ultrasonic thickness readings on each floor plate.	• Note: The random UT thickness readings and prove-up of the MFE indications revealed the floor to be 0.250". Several of the sketch plates had a thickness between 0.255" to 0.271". For inspection purposes 0.250" was used for the calculations and pit gauging. • Floor thickness: (0.250") / Sketch thickness: (0.250")
7	Locate unacceptable voids under the floor plates.	• Several minor-to-moderate floor plate voids were noted on the sketch and floor plates. The voids have depressions of less than one inch (1") and bottom out on the foundation below. Recommend reevaluating at the next out-of-service interval.
8	Indicate the number and condition of sumps including coating. Conduct Ultrasonic Thickness readings on sump.	• Acceptable – the tank has two (2) welded sumps. The sumps are in good condition with no weld corrosion or pitting noted. • Diameter of sumps: (24") / Height of sumps: (18") • The two (2) sump bottoms (welded) have a nominal thickness of 0.1875" (3/16"). The sump walls are 0.375" nominal thickness. No corrosion was noted inside the sumps. Note: This is not a recommended practice to have sump bottoms with this thickness. Consideration should be given to replacing the sump bottoms with thicker plate.

BOTTOM INSPECTION (Cont.)

Item	Description	Comments
9	Any areas of edge settlement, bulges or low areas that don't drain.	• Note: No edge settlement noted.
10	Conduct visual inspection of bottom / bottom coating. Record all findings.	• Note: The tank bottom has a thick epoxy coating on the lap rivet seams (6" each side). The epoxy has varying thicknesses and is broken-up or completely missing in numerous areas. Also, the floor plates are coated with a very thin film primer coat (approximately 5 mils). The primer coat is in good condition overall with failure areas noted at the seams.
11	Are there striker plates under the vacuum breakers and floating roof legs?	• N/A – Fixed roof only.
12	Visually inspect roof support columns and lateral clips.	• The nineteen (19) fixed roof columns sit on the floor with no bearing plates or lateral retaining clips. Recommend installing seal welded bearing plates, under the column base pads (10"x 13") and installing lateral retaining clips. Note: The outer row of columns (12 total) have three I4" beams, welded to the bottom of the base pads. This is not a recommended practice. • The fixed roof column at floor plate #78 has a minor twist. It is believed the column can be corrected, when the bearing plate and lateral clips are installed. • Type of Supports: Channel Column • Number of Columns: Nineteen (19) • Size of Column: C9"x2.5" / C7"x2" riveted at 90° • Size of Base: 10"x 13" plate (riveted to columns) • Size of Base Plate: None
13	Conduct visual inspection of reinforcements. Do all piping supports have support pads?	• The heating coil piping supports have 5" diameter support pads, that are not seal welded (stitch welded only). It is highly recommended that the support pads (165 total) be seal welded, before installing the new floor liner. Note: Several areas of significant topside corrosion were noted extending under the pads. • Recommend seal welding the 8" diameter support pad for the temperature probe support (floor plate #22). • The 2" pipe support for the 8" truck suction pipe (plate #35) is welded to the floor and pipe, allowing no movement. Recommend modifying the support to allow movement of the piping and installing a seal welded support pad on the floor. Also, recommend installing a product drainage notch, in the bottom of the support.
14	Comments on possible ways to increase product storage space.	• N/A – No comments.

INTERVAL TO NEXT BOTTOM INSPECTION:

$$O_r = \frac{(\text{min}RT_{bc} \text{ or } RT_{ip}) - MRT}{S_tP_r + UP_r}$$

Nominal Bottom Plate Thickness (inches)	0.250”	
Maximum Internal Pit Depth	0.230”	
Maximum Underside Pit Depth	0.090”	
Year of Inspection (Yi)	2011	
Year of Construction (Yo)	1936	
Internal Corrosion Rate (StPr)	0.0017”	
Underside Corrosion Rate (UPr)	0.0013”	
Minimum Remaining Plate Thickness after Repairs for Underside (RT_{bc})	0.150”	(or MFE threshold)
Minimum Remaining Plate Thickness after Repairs for Internal (RT_{ip})	0.120”	(or MFE threshold)
Minimum Remaining Thickness, At the Next Inspection (MRT)	0.100”	

Interval to next bottom inspection	**6.67 Years** (RRT – 0.10) / (Int + US C.R.)
If a thin film coating is installed after repairs (Expected life of coating must equal or exceed this interval) (Internal Corrosion Rate = 0)	**15.38 Years** (RRT – 0.10) / (0 + US C.R.)
If a reinforced fiberglass liner is installed after repairs (Expected life of coating must equal or exceed this interval) (Internal Corrosion Rate = 0) (MRT @ next inspection = 0.05 inches)	**20 Years Max.** (RRT – 0.05) / (0 + US C.R.)
If cathodic protection is operating effectively (Underside corrosion Rate = 0)	**11.76 Years** (RRT – 0.10) / (Int + 0 C.R.)

BOTTOM PLATE REDUCTION TABLE

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
2A	.13-.14	0.110"-0.120"	21-34	09-19	X					PITTING (3)
B	.13	0.120"	58	06	X					PIT
C	.13	0.120"	82	06	X					PIT
D	.13-.14	0.110"-0.120"	122-144	09-18	X					PITTING (3)
4A	.13	0.120"	62	32	X					PIT
B	.13	0.120"	64	09	X					PIT
5A	.13	0.120"	40	63	X					PIT
B	.14	0.110"	191	60	X					PIT
C	.13	0.120"	185	42	X					PIT
D	.14	0.110"	205	29	X					PIT
6A	.13-.15	0.100"-0.120"	06-77	37-63	X					PITTING (16)
B	.13	0.120"	52	11	X					PIT
C	.13	0.120"	94	64	X					PIT
D	.13	0.120"	112	58	X					PIT
E	.13	0.120"	128-131	61-64	X					PIT
F	.13-.14	0.110"-0.120"	151-156	62	X					PITTING (2)
8A	.13	0.120"	72	33	X					PIT
11A	.18	0.070"	138	18	X					PIT
B	.13	0.120"	150	65	X					PIT
C	.13	0.120"	180	62	X					PIT
D	.13	0.120"	214	64	X					PIT
E	.13	0.120"	227	66	X					PIT
F	.13	0.120"	236	62	X					PIT
G	.13	0.120"	235	44	X					PIT
H	.13	0.120"	233	24	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
13A	.14	0.110"	82	58	X					PIT
14A	.10	0.050"	09	09	X					PIT
B	.14	0.110"	51	63	X					PIT
C	.13	0.120"	224	35	X					PIT
D	.13	0.120"	243	42	X					PIT
E	.13-.14	0.110"-0.120"	230-243	55-65	X					PITTING (3)
15A	.16	0.090"	194	15	X					PIT
16A	.13	0.120"	62	57	X					PIT
B	.15	0.100"	132	10	X					PIT
17A	-	0.190"	158	38		X				1 INDICATION
24A	.15	0.100"	10	04	X					PIT
B	.13	0.120"	12	20	X					PIT
C	.13	0.120"	54	60	X					PIT
D	.13	0.120"	71	56	X					PIT
25A	.14	0.110"	129	23	X					PIT
B	.13	0.120"	160	35	X					PIT
C	.13	0.120"	227	16	X					PIT
D	.14	0.110"	243	20	X					PIT
E	.13	0.120"	242	30	X					PIT
F	.13-.14	0.110"-0.120"	234-241	53-58	X					PITTING (2)
26A	.14	0.110"	66	25	X					PIT
B	.13	0.120"	92	39	X					PIT
C	.13	0.120"	100	32	X					PIT
D	.15	0.100"	116	36	X					PIT
E	.14	0.110"	136	27	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
26F	.13	0.120"	200	38	X					PIT
G	.13	0.120"	220	33	X					PIT
28A	.14	0.110"	30-38	42-46	X					PITTING (2)
B	.13	0.120"	88	12	X					PIT
29A	.16	0.090"	47	13	X					PIT
B	.13	0.120"	66	31	X					PIT
C	.10-.16	0.090"-0.150"	66-80	46-60	X					PITTING (6)
D	.13	0.120"	114	59	X					PIT
E	.15	0.100"	142	21	X					PIT
F	.13	0.120"	168	53	X					PIT
30A	-	0.161"-0.183"	27-30	37-42		X				2 INDICATIONS
31A	.13	0.120"	57	68	X			X		PIT
B	.15	0.100"	11	08	X			X		PIT
32A	.13-.15	0.100"-0.120"	08-14	54-55	X			X		PITTING (2)
33A	.14	0.110"	05	08	X					PIT
B	.13	0.120"	49	62	X					PIT
C	.14	0.110"	141	10	X					PIT
D	.13	0.120"	226	30	X					PIT
34A	.14	0.110"	149	66	X					PIT
B	.14	0.110"	215	69	X					PIT
35A	.15	0.100"	33	57	X					PIT
B	.13	0.120"	53	13	X					PIT
C	.13-.14	0.110"-0.120"	69-74	09-13	X					PITTING (2)
D	.13	0.120"	76	49	X					PIT
E	.14	0.110"	91	15	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
35F	.13	0.120"	109	15	X					PIT
36A	.13	0.120"	09	29	X					PIT
B	.13	0.120"	36	20	X					PIT
37A	.14	0.110"	10	44	X					PIT
B	.14	0.110"	170	21	X					PIT
C	.14	0.110"	184	65	X					PIT
D	.13	0.120"	215	43	X					PIT
E	.13	0.120"	225	61-68	X					PITTING (2)
38A	.13	0.120"	44	50	X					PIT
B	.15	0.100"	56	52	X					PIT
C	.13	0.120"	151	21	X					PIT
D	.13	0.120"	162	33	X					PIT
39A	.14	0.110"	171	48	X					PIT
B	.13	0.120"	174-189	59-65	X					PITTING (3)
C	.13-.15	0.100"-0.120"	184-187	37-46	X					PITTING (2)
D	.13	0.120"	182	13	X					PIT
E	.13	0.120"	194	16	X					PIT
40A	.19	0.060"	130	42	X					PIT
41A	-	0.160"-0.190"	24-33	50-63		X				4 INDICATIONS
B	.14	0.110"	154	37	X					PIT
C	.13	0.120"	199	48	X					PIT
D	.13	0.120"	208	19	X					PIT
44A	.13	0.120"	148	68	X					PIT
45A	.14	0.110"	49	32	X					PIT
B	.13	0.120"	57	23	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
45C	.14-.16	0.090"-0.110"	95-102	18-27	X					PITTING (3)
D	.13	0.120"	117	52	X					PIT
E	.13	0.120"	164	09	X					PIT
F	.16	0.090"	167	49	X					PIT
G	.15	0.100"	181	36	X					PIT
H	.14	0.110"	212	61	X					PIT
46A	.13	0.120"	57	52	X					PIT
B	.13	0.120"	129	23	X					PIT
C	.13-.15	0.100"-0.120"	149	20	X					PIT
D	.14	0.110"	170	30	X					PIT
E	.13	0.120"	176	53	X					PIT
F	.13	0.120"	178-184	40-41	X					PIT
47A	.13-.15	0.100"-0.120"	16-26	04-17	X					PITTING (4)
B	.13	0.120"	24	45	X					PIT
C	.13	0.120"	94	21	X					PIT
D	.14	0.110"	108	16	X					PIT
E	.13	0.120"	112	56	X					PIT
F	.13	0.120"	150	50	X					PIT
G	.13	0.120"	194	32	X					PIT
48A	.13	0.120"	06	19	X			X		PIT
49A	.13	0.120"	49	52	X					PIT
B	.13	0.120"	54	12	X					PIT
51A	.13	0.120"	13	34	X					PIT
B	.13	0.120"	13	56	X					PIT
C	.23	0.020"	97	53	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
51D	.13	0.120"	110	29	X					PIT
E	.13	0.120"	175	07	X					PIT
F	.13	0.120"	222	42	X					PIT
G	.13	0.120"	234	28	X					PIT
52A	.13	0.120"	08	42	X					PIT
B	.13	0.120"	19	41	X					PIT
C	.13-.15	0.100"-0.120"	29-49	43-50	X					PITTING (4)
D	.16	0.090"	51	26	X					PIT
E	.13	0.120"	52	06	X					PIT
F	.13	0.120"	72-75	16-20	X					PITTING (2)
G	.13	0.120"	116	60	X					PIT
H	.13	0.120"	178	38	X					PIT
I	.13	0.120"	186	04	X					PIT
53A	.13	0.120"	165	17	X					PIT
B	.14	0.110"	175	24	X					PIT
C	.13	0.120"	185	11	X					PIT
D	.13	0.120"	211	16	X					PIT
E	.13	0.120"	225	09	X					PIT
F	.13	0.120"	243	46	X					PIT
54A	.21	0.040"	08	32	X					PIT
B	.15	0.100"	86	26	X					PIT
C	.19	0.060"	93	34-37	X					PIT
D	.13	0.120"	198	26	X					PIT
55A	.13	0.120"	38	65	X			X		PIT
57A	.13	0.120"	10	49	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
58A	.13	0.120"	18	59	X					PIT
B	.14	0.110"	46	62	X					PIT
C	.13	0.120"	103	17	X					PIT
D	.15	0.100"	99	62	X					PIT
E	.13	0.120"	109	64	X					PIT
F	.13	0.120"	222	08	X					PIT
G	.16	0.090"	222	24	X					PIT
H	.13	0.120"	227	30-34	X					PIT
I	.13	0.120"	240	33	X					PIT
59A	.15	0.100"	60	25	X					PIT
B	.13	0.120"	74	40	X					PIT
C	.14	0.110"	85	28	X					PIT
D	.15	0.100"	84	10	X					PIT
E	.13	0.120"	165	26	X					PIT
F	.18	0.070"	200	26	X					PIT
G	.13	0.120"	207	14	X					PIT
60A	.10-.16	0.090"-0.150"	06-36	28-64	X					PITTING (9)
B	.13-.14	0.110"-0.120"	99-108	10-20	X					PITTING (3)
C	.22	0.030"	121	63	X					PIT
D	.20	0.050"	151	27	X					PIT
E	.17	0.080"	170	61	X					PIT
F	.14	0.110"	184	50	X					PIT
G	.13-.15	0.100"-0.120"	203-214	42-51	X					PITTING (4)
61A	.14	0.110"	44	45	X					PIT
B	.16	0.090"	50-56	34-42	X					PITTING (3)

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
61C	.13	0.120"	152	42	X					PIT
62A	.13	0.120"	07	62	X					PIT
B	.13	0.120"	54	29	X					PIT
C	.13	0.120"	65	19	X					PIT
D	.13	0.120"	45	06	X					PIT
63A	.13	0.120"	22	45	X					PIT
B	.14	0.110"	42	42	X					PIT
C	.13	0.120"	40	21	X					PIT
D	.13	0.120"	50	27	X					PIT
E	.13	0.120"	109	12	X					PIT
F	.14	0.110"	109	23	X					PIT
G	.14	0.110"	144	28	X					PIT
H	.13	0.120"	167	22	X					PIT
I	.16	0.090"	210	41-45	X					PITTING (2)
64A	.13	0.120"	13	23	X					PIT
B	.13	0.120"	21	14	X					PIT
C	.13	0.120"	42	38	X					PIT
D	.13	0.120"	49	49	X					PIT
E	.13	0.120"	51	39	X					PIT
F	.13	0.120"	79	40	X					PIT
G	.13	0.120"	91	38	X					PIT
H	.14	0.110"	89	12	X					PIT
I	.15	0.100"	98-105	42-43	X					PITTING (2)
J	.15	0.100"	122	33	X					PIT
K	.13	0.120"	173	42	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
64L	.13-.14	0.110"-0.120"	231-235	48-50	X					PITTING (2)
65A	.14	0.110"	48	60	X					PIT
B	.17	0.080"	155	59	X					PIT
C	.13	0.120"	244	47	X					PIT
66A	.16	0.090"	53	65	X					PIT
B	.13	0.120"	200	58	X					PIT
C	.14	0.110"	210	56	X					PIT
67A	.13	0.120"	112	62-66	X					PITTING (2)
B	.14	0.110"	139	43	X					PIT
C	.14	0.110"	156	63	X					PIT
D	.13	0.120"	210	04	X					PIT
E	.13	0.120"	238	63	X					PIT
70A	.14	0.110"	10	33	X					PIT
B	.15	0.100"	16	26	X					PIT
C	.13	0.120"	76	65	X					PIT
71A	.13	0.120"	35-40	35-37	X					PITTING (2)
B	.13	0.120"	76	50	X					PIT
C	.13	0.120"	134	47	X					PIT
D	.15	0.100"	234	02	X					PIT
72A	.13	0.120"	10	34	X					PIT
B	.13	0.120"	16	58	X					PIT
C	.15	0.100"	22-29	27-30	X					PIT
D	.13	0.120"	34	21	X					PIT
E	.14	0.110"	54	25	X					PIT
F	.14	0.110"	63	24	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
72G	.13	0.120"	79	10	X					PIT
H	.15	0.100"	103-111	30	X					PITTING (3)
I	.16	0.090"	112	58	X					PIT
J	.13	0.120"	199	59	X					PIT
73A	.14	0.110"	20	51	X					PIT
B	.13	0.120"	40	54	X					PIT
C	.15	0.100"	50	11	X					PIT
D	.15	0.100"	75	64	X					PIT
74A	.14	0.110"	72	27	X					PIT
B	.14	0.110"	78	04	X					PIT
C	.17	0.080"	126	33	X					PIT
D	.15	0.100"	133	34	X					PIT
E	.14	0.110"	139	57	X					PIT
F	.13	0.120"	208-215	12-16	X					PITTING (3)
G	.13	0.120"	224	08	X					PIT
H	.14	0.110"	224	29	X					PIT
I	.13	0.120"	216	47	X					PIT
J	.14	0.110"	220	55	X					PIT
K	.13	0.120"	160	32	X					PIT
L	.13	0.120"	182	28	X					PIT
77A	.13	0.120"	21	16	X					PIT
B	.13	0.120"	103	25	X					PIT
C	.13	0.120"	101-108	53	X					PIT
D	.13	0.120"	164	46	X					PIT
78A	.13	0.120"	58	59	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
80A	.13	0.120"	22	42	X					PIT
B	.13	0.120"	40	26	X					PIT
C	.13	0.120"	71	64	X					PIT
D	.13	0.120"	104	09	X					PIT
81A	.13	0.120"	91	43	X					PIT
82A	.14	0.110"	76	60	X					PIT
B	.13	0.120"	68	07	X					PIT
84A	.13	0.120"	27	32	X					PIT
B	.13	0.120"	140	08	X					PIT
87A	.13	0.120"	137	13	X			X		PIT
B	.13	0.120"	112	64	X			X		PIT
C	.13	0.120"	13	13	X			X		PIT
88A	.13	0.120"	41	44	X					PIT
89A	.13	0.120"	96	44	X					PIT
B	.13	0.120"	110	25	X					PIT
C	.15	0.100"	172	13	X					PIT
D	.13	0.120"	188	55	X					PIT
E	.13-.16	0.090"-0.120"	192-210	17-25	X					PITTING (3)
F	.13-.15	0.100"-0.120"	208-214	04-08	X					PITTING (2)
G	.13	0.120"	226	26	X					PIT
H	.14	0.110"	227	15	X					PIT
90A	.13	0.120"	28	03	X					PIT
B	.13	0.120"	33	58	X					PIT
C	.14	0.110"	78	10	X					PIT
D	.13	0.120"	151	57	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
91A	.13-.15	0.100"-0.120"	17-48	08-33	X					PITTING (11)
B	.15	0.100"	96	25	X					PIT
C	.14	0.110"	139	57-61	X					PITTING
D	.13	0.120"	182	23	X					PIT
92A	.13-.15	0.100"-0.120"	117-145	47-57	X					PITTING (4)
B	.13	0.120"	165	60	X					PIT
C	.13	0.120"	176	63	X					PIT
D	.16	0.090"	211	48	X					PIT
E	.13	0.120"	243	42	X					PIT
94A	.13	0.120"	67	07	X					PIT
95A	.13-.14	0.110"-0.120"	07-12	54-57	X					PITTING (2)
B	.13	0.120"	33	43	X					PIT
C	.13	0.120"	52	15	X					PIT
D	.13	0.120"	102	45	X					PIT
E	.16	0.090"	115	07	X					PIT
F	.13-.15	0.100"-0.120"	124-147	10-32	X					PITTING (7)
G	.13-.15	0.100"-0.120"	130-136	60-62	X					PITTING
H	.13	0.120"	141	49	X					PIT
I	.14	0.110"	166	25	X					PIT
J	.13	0.120"	183-185	52-62	X					PITTING (4)
K	.14	0.110"	217	10	X					PIT
L	.13	0.120"	215	32	X					PIT
96A	.14	0.110"	74	61	X					PIT
B	.15	0.100"	83	15	X					PIT
C	.13	0.120"	114	32	X					PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
96D	.13	0.120"	136	10	X					PIT
E	.13	0.120"	156	42	X					PIT
F	.13	0.120"	235	49	X					PIT
G	.13	0.120"	229	34	X					PIT
97A	.18	0.070"	48	11	X					PIT
B	.13	0.120"	22	51	X			X		PIT
98A	.13	0.120"	69	36	X					PIT
100A	.13	0.120"	46	51	X			X		PIT
101A	.13	0.120"	24	49	X					PIT
B	.13	0.120"	26	24	X					PIT
C	.13	0.120"	37	09	X					PIT
D	.13	0.120"	74	21	X					PIT
E	.13	0.120"	124	55	X					PIT
102A	.15	0.100"	155	55	X					PIT
B	.15	0.100"	48-54	37-40	X					PIT
C	.13-.14	0.110"-0.120"	32-39	49-57	X					PIT (2)
103A	.13	0.120"	35-42	60	X			X		PIT
B	.13	0.120"	57	45	X			X		PIT
C	.14	0.110"	44	31	X			X		PIT
D	.16	0.090"	63-68	17-19	X			X		PITTING
E	.15	0.100"	93	21	X			X		PIT
F	.14	0.110"	84-86	16	X			X		PIT
G	.13	0.120"	97	08-13	X			X		PIT
104A	.13	0.120"	54	39	X			X		PIT
B	.14	0.110"	27	39	X			X		PIT

BOTTOM PLATE REDUCTION TABLE (Cont.)

Plate No.	Depth	Remaining Plate	"X" Location	"Y" Location	Internal	Underside	Critical Zone	N/E Corner	Mech.	Comments
104C	.14	0.110"	26	32	X			X		PIT
105A	.15	0.100"	33	16	X					PIT
B	.18	0.070"	24	40	X					PIT
C	.13	0.120"	60	23	X					PIT
D	.14	0.110"	117	28	X					PIT
E	.13	0.120"	119	48	X					PIT
F	.13	0.120"	203-208	05-08	X					PITTING (2)
106A	.13	0.120"	118	41	X					PIT
B	.13	0.120"	167	37	X					PIT
107A	.13	0.120"	18	06	X			X		PIT
109A	.13	0.120"	15	15	X					PIT

SHELL INSPECTION

Item	Description	Comments
1	Inspect shell ladders, stairways, platforms and walkways.	• The radial stairway is in good structural condition with minor coating failure. Recommend monitoring and recoat the stairway when conditions warrant. Note: The vertical ladder (on the internal and external shell) has no fall cage. Consideration should be given to installing a fall cage on the ladder, if the ladder is to be used.
2	Inspect shell for significant distortions, creases and buckling.	• Acceptable – No significant distortions noted from inside the tank. Limited exam, due to the external insulation.
3	Visually inspect shell for out-of-roundness.	• Acceptable – No visual out-of-roundness noted.
4	Inspect shell support welds for corrosion or damage.	• N/A – No exam, due to the external insulation.
5	Inspect windgirder for corrosion or damage.	• N/A – No windgirder or stiffener ring.
6	Inspect external shell for leaks or significant corrosion areas.	• Dried product is noted on the shell plates on the west side of the tank, where the insulation was removed (3'x 3' area). It is believed, the product is weeping from the riveted shell seams. Recommend performing a visual exam, if the insulation is ever removed.
7	Inspect exterior shell-to-bottom weld area for corrosion, defects or thinning of the chime bottom plate.	• N/A – The external floor-to-shell tub area is covered by the insulation and is below the grade. The internal floor-to-shell rivet angle is in good condition with no significant defects or corrosion noted.
8	Inspect shell coating for failure, peeling or fading. Overall condition:	• The visual exam of the shell coating was very limited, due to the insulation. However, the coating has hairline cracking, fading and peeling at the inspection ports and areas where the insulation was cut and removed. • Fair-to-poor
9	Inspect both the internal and external shell for pitting, corrosion, rubbing or grooving.	• Note: No significant internal corrosion or pitting noted. Also, no significant thinning was noted at the inspection ports.

SHELL INSPECTION (Cont.)

Item	Description	Comments
10	Inspect horizontal and vertical seams for corrosion. Note weld or rivet joint details.	• The tank has seven (7) butt and lap riveted shell courses. Note: The shell course configurations are as follows: • 1st and 2nd courses: 4 row / butt riveted with backing plates • 3rd course: 4 row / lap riveted • 4th and 5th courses: 3 row / lap riveted • 6th and 7th courses: 2 row / lap riveted • Horizontal seams are single row lap riveted
11	Inspect shell overflows for corrosion and screening.	• N/A – No shell vents or overflows.
12	Inspect shell for any lap patches.	• Acceptable – No internal shell lap patches noted.
13	Is the tank grounded?	• The tank is not grounded. Consideration should be given to grounding the tank.
14	Does the tank have a legible data plate?	• No ID / data placard is visible on the tank. Consideration should be given to installing a data placard on the tank. Note: The NFPA (hazard) placards are in good visible condition.
15	Note if supports have reinforcement pads welded to the shell.	• N/A – No exam, due to the external insulation.
16	Conduct visual inspection of shell insulation. Inspect for damage.	• The shell insulation is in fair condition overall. The insulation is distorted / damaged, around the heating coils on the northwest side of the tank. Also, the insulation and shell were wet on the top three (3) courses (at the inspection ports). Recommend sealing the insulation angle at the top of the tank, when the roof is replaced. Note: Recommend sealing all insulation penetrations with caulking (nozzles / support angles) when all repairs are complete. • Recommend a complete visual exam of the external shell, rivets, nozzles and repads, if the insulation is ever removed. The visual exam was very limited, at the time of the inspection.

INTERVAL TO NEXT SHELL INSPECTION:

Based upon the current average shell plate thickness and an assumed general corrosion rate of 0.002” per year, the required interval to the next shell testing and inspection (in-service) is five (5) years.

SHELL THICKNESS EVALUATION

Shell Plate Thickness and Safe Fill Height Calculations

The following calculations determine the minimum acceptable shell plate thickness (t(min)) and safe fill heights (product and water) for tanks equal to or less than 200 feet in diameter in accordance with Section 2 of API 653.

Minimum Acceptable Shell Plate Thickness [t(min)]

t(Min.) = 2.6 D (H - 1) G / S E

Where:

t(min) = Minimum acceptable shell plate thickness (in.)

H = Product Height (ft.)
D = Tank Diameter (ft.)
G = Specific Gravity
S = Allowable stress (psi)
E = Joint Efficiency
*CA = Corrosion Allowance (in.)

* Corrosion rate (mils per year) times the number of years to the next shell inspection. When corrosion rates are unknown, CA = 0.002 inches per year x 5 years = 0.010.

Field Measured Shell Thickness

t(ave) = Average UT readings in uncorroded areas
t1 = Lowest average thickness in "pitted area"
t2 = Minimum thickness in "pitted area" (exclusive of isolated pits)
tp = Remaining shell thickness at an isolated pit

The following four (4) checks must be made for each shell course:

Check No.	Measured Thickness*	Check	Calculated Thickness Limit
1	t(ave)	> =	t(LV) = t(min) + CA
2	t1	> =	t(LV) = t(min) + CA
3	t2	> =	t(LV) = 0.6 x t(min) + CA
4	tp	> =	t(LV) = 0.5 x t(min)

* All tanks will have a t(ave) for each course, but many tanks will have no measurements of t1, t2 or tp in any particular course if these conditions do not exist.

SHELL THICKNESS EVALUATION (Continued)

If any check for any shell course fails, there are three (3) options:

1. Retire the tank from service.
2. Repair the area causing the failure so that it passes.
3. Limit the safe fill height (H) to reduce t(LV) to allow the area to pass.

Tank Height (ft):	42.00
D (ft):	117.00
SG (Water):	1.0
SG (Used):	1.00

Shell Thickness Variables

Course	Course Height (ft)	H (ft)	Allowable Stress S (Psi)	Joint Efficiency E	CA (in)
1	6.67	42.00	21,000	0.90	0.010
2	6.67	35.33	21,000	0.90	0.010
3	6.67	28.66	21,000	0.75	0.010
4	5.50	21.99	21,000	0.70	0.010
5	5.50	16.49	21,000	0.70	0.010
6	5.50	10.99	21,000	0.60	0.010
7	5.49	6.00	21,000	0.60	0.010

	Check #1			Check #2			Check #3			Check #4		
Course	t(ave) (in)	t(LV)* (in)	** P/F	t1 (in)	t(LV)* (in)	** P/F	t2 (in)	t(LV)* (in)	** P/F	tp (in)	t(LV)* (in)	** P/F
1	0.583	0.670	F	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
2	0.482	0.563	F	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
3	0.478	0.544	F	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
4	0.406	0.444	F	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
5	0.308	0.331	F	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
6	0.252	0.251	P	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--
7	0.251	0.131	P	N/A	N/A	--	N/A	N/A	--	N/A	N/A	--

*t(LV) - Limiting thickness differs for each check. See previous page.
**P indicates course passes. F indicates course fails
N/A indicates that this condition was not found on this course.

SHELL THICKNESS EVALUATION (Continued)

	Check #1		Check #2		Check #3		Check #4	
Course	Water	SG 1.0	Water	SG 1.0	Water	SG 1.0	Water	SG 1.0
1	36.60	36.60	N/A	N/A	N/A	N/A	N/A	N/A
2	37.00	37.00	N/A	N/A	N/A	N/A	N/A	N/A
3	38.57	38.57	N/A	N/A	N/A	N/A	N/A	N/A
4	40.15	40.15	N/A	N/A	N/A	N/A	N/A	N/A
5	40.91	40.91	N/A	N/A	N/A	N/A	N/A	N/A
6	NR	NR	N/A	N/A	N/A	N/A	N/A	N/A
7	NR	NR	N/A	N/A	N/A	N/A	N/A	N/A

Controlling Safe Fill Height	
Fluid	Height (ft)
Water	36.60
SG 1.0	36.60

NR - Safe Fill Height is not restricted.
N/A - Condition does not apply to this course.

SHELL PLATE ULTRASONIC THICKNESS READINGS

Reading	Course 1	Course 2	Course 3	Course 4	Course 5	Course 6	Course 7
1	.589	.486	.475	.404	.301	.240	.256
2	.581	.478	.481	.407	.308	.247	.250
3	.580	.481	.478	.407	.314	.269	.246
Average:	**0.583”**	**0.482”**	**0.478”**	**0.406”**	**0.308”**	**0.252”**	**0.251”**

SHELL NOZZLES AND APPURTENANCES INSPECTION

Item	Description	Comments
1	Inspect nozzle flanges and bolting for leaks and corrosion.	• Acceptable – No flange leaks noted.
2	Inspect nozzles for pipe strain or misalignment.	• Acceptable – No pipe stain or misalignment noted.
3	Inspect valves for leaks or damage.	• Note: The two (2) waterdraw nozzles have typical gate valves. • Inlet nozzle "X" (16") has an older design gate valve. The valve appears to be in good condition. However, consideration should be given to installing a newer design gate valve, while the tank is out-of-service. • The valve for suction nozzle "I" has a heavy product stain at the valve stem. Recommend monitoring or repair while the tank is out-of-service.
4	Do nozzles, which require pressure relief, have crossovers at the valve?	• The pressure relief piping for suction nozzle "I" (at pop-off) has a heavy product stain at the threads. Recommend repairing the piping while the tank is out-of-service. Note: The pressure relief system for nozzle "I" has the older design crossover piping around the valve. Consideration should be given to installing a typical TransMontaigne design system, which empties back into the tank with a 2" shell nozzle. • Inlet nozzles "J" (10") & "X" (16") and truck suction nozzle "Q" (8") have no pressure relief systems at the valves. Recommend installing typical TransMontaigne design pressure relief systems at these nozzles.
5	Inspect mixer, mixer gaskets for leakage and defects. Are mixer housings free of debris / nest?	• Note: The mixer was removed from manway "A". Minor product stains were noted at the swivel. Recommend monitoring the mixer when reinstalled. Also, no independent grounding cable was noted at the manway. Consideration should be given to installing a grounding cable on the mixer, when reinstalled. • N/A – Closed motor design.
6	Verify operation of gauging system. Type of Gauges	• The Varec float has been removed from the tank. Also, the Varec gauge housing is holding water and is not legible. Recommend repairing the Varec float gauge, before returning the tank to service. • Varec 2500 ("B") / ENRAF Smart Radar
7	Inspect manways, nozzles and attachments for signs of corrosion or defects.	• Nozzle "C" (8" / not in use) has a neck thickness (0.313"), less than recommended by API-650. • Inlet nozzle "J" (10") has a neck thickness (0.378"), less than recommended by API-650. • Truck suction nozzle "Q" (8") has a neck thickness (0.312"), less than recommended by API-650. • Inlet nozzle "X" (16") has a neck thickness (0.373"), less than recommended by API-650. • Waterdraw nozzle "AA" (2") has a neck thickness (0.145"), less than recommended by API-650.

SHELL NOZZLES AND APPURTENANCES INSPECTION (Cont.)

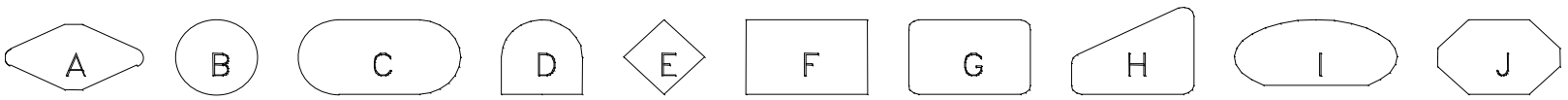
Item	Description	Comments
7 (Cont.)	Inspect manways, nozzles and attachments for signs of corrosion or defects.	• Butt inserts "F" and "Z" (8") have dimensions, less than recommended by API-653. Note: Inserts were viewed from inside the tank. • Low suction nozzle "H" (10") still has the 90° elbow (hard piping) in place. The low suction piping is welded to the 16" suction piping (nozzle "I") with no valve. Recommend removing the 90° low suction elbow and installing a typical TransMontaigne design removable hose. • Temperature probe couplets "M" (2nd course), "N" (4th course), "O" (6th course) and "W" (1st course) are no longer being used. Consideration should be given to removing the couplets and internal piping, if no longer needed. • Consideration should be given to recoating all nozzle and manway necks, flanges and covers that are not insulated. These appurtenances have coating failure with active rust discoloration. • Doorsheet "B" has square (non-rounded) corners. API-653 recommends rounded corners on butt inserts / doorsheets. Note: The doorsheet was viewed from inside the tank.
8	Do all nozzle reinforcement pads have tell-tale hole? Nozzle reinforcement pads with plugged tell-tale holes.	• N/A – No exam (Insulated) • N/A – No exam (Insulated)
9	Verify adequate reinforcement around nozzles and appurtenances.	• N/A – No exam (Insulated)
10	Verify adequate nozzle and appurtenances weld spacing (horizontal and vertical weld seams and distance between reinforcement pads).	• N/A – No exam, due to the insulation. Also, the shell is riveted.
11	Verify interior nozzles have been back welded to the shell.	• Suction nozzle "I" (16") was installed by the "Hot-Tap" method. The internal nozzle neck is not back welded to the shell. API-653 recommends "Hot-Taps" larger than 10" in diameter, be back welded internally. Recommend back welding before returning the tank to service.
12	Check reinforcement pads for adequate fillet welds around pads.	• N/A – No exam, due to the insulation.
13	Visually inspect welds for defects on all nozzles and appurtenances.	• The 8" piping at truck suction nozzle "Q" has significant weld corrosion and coating failure at the tie-in. The area is located between the valves, 4' from the tank and is marked with a yellow paint marker. Consideration should be given to touching-up the weld in this area.

SHELL NOZZLES AND APPURTENANCES INSPECTION (Cont.)

Item	Description	Comments
14	Does the inlet nozzle have a diffuser? Describe the diffuser. Does the suction nozzle have an internal trough?	• Note: The inlet diffuser at nozzle "X" (16" / no flange) is 9'-3" long with slots the entire length. The slots are 16"L x 2"W (8" apart). The diffuser has a "U" bolt piping support with pads and is in good structural condition. Inlet nozzle "J" (10" / riveted) is flush with the shell and open ended. Consideration should be given to installing a diffuser at this nozzle, if needed. • Size of line: 16" • Suction nozzle "I" (16" / "hot-tap") has no type of suction trough. Consideration should be given to installing a typical TransMontaigne design suction trough, if needed. Note: The low suction piping is a 90° elbow-down design, which sits approximately 12" above the floor. • Size of line: 16"
15	Check welds on shell mounted davit clips above large valves or equipment.	• Note: The mixer is installed in the cover of manway "A" and has no type of supports. Consideration should be given to supporting the mixer from the shell.
16	Check floating suction for damage and proper support. Check condition of shell mounted angle limiting brackets or angle limiting chains mounted to bottom.	• N/A – No floating suction.

SHELL NOZZLE AND APPURTENANCES TABLE

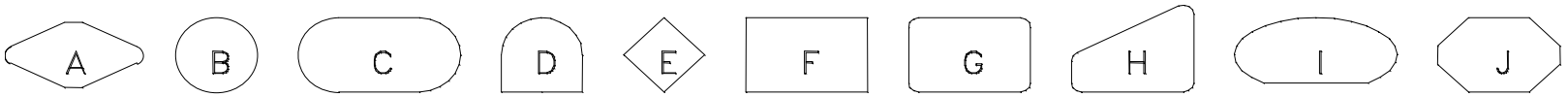
Item	Description	Size	Location	CL Elev.	Repad Shape	Repad Height	Repad Width	Repad Thk.	Neck Thk.	Flange Thk.	Cover Thk.	Weld Space	Tell Tale	Comments
A	Manway	24”	0’-0”	36”	?	?	?	?	.384	.478	1.024	?	?	w/ mixer (removed)
B	Doorsheet	60”x60”	11’-8”	39.5”	?	?	?	?	.589	-	-	?	?	
C	Nozzle	8”	32’-3”	43”	?	?	?	?	.313	1.125	1.270	?	?	Not in use
D	Doorsheet	49”x49”	40’-0”	40”	?	?	?	?	.620	-	-	?	?	
E	Couplet	2”	90’-9”	5.5”	?	?	?	?	-	-	-	?	?	Plugged
F	Butt Insert	8”	95’-0”	8”	?	?	?	?	.463	-	-	?	?	
G	Nozzle	2”	97’-2”	12.5”	?	?	?	?	.229	.781	-	?	?	Waterdraw
H	Nozzle	10”	101’-3”	25.5”	?	?	?	?	.404	1.171	-	?	?	* / Low suction
I	Nozzle	16”	105’-1”	36”	?	?	?	?	.497	1.447	-	?	?	Suction (Hot Tap)
J	Nozzle	10”	121’-7”	21”	?	?	?	?	.378	1.233	-	?	?	* / Inlet
K	Tank Gauge	-	168’-3”	-	?	?	?	?	-	-	-	?	?	Varec (not in use)
L	Ladder	-	170’-7”	-	?	?	?	?	-	-	-	?	?	Vertical (inside & outside)
M	Nozzle	1.5”	172’-3”	-	?	?	?	?	-	-	-	?	?	Temp. probe / 2nd course
N	Nozzle	1.5”	172’-3”	-	?	?	?	?	-	-	-	?	?	Temp. probe / 4th course
O	Nozzle	1.5”	172’-3”	-	?	?	?	?	-	-	-	?	?	Temp. probe / 6th course
P	Manway	20”	183’-9”	31.5”	?	?	?	?	.492	.571	.608	?	?	*
Q	Nozzle	8”	221’-8”	49”	?	?	?	?	.312	1.156	-	?	?	Suction for trucks
R	Nozzle	2”	250’-1”	19”	?	?	?	?	.193	.618	-	?	?	Heating coil
S	Nozzle	2”	251’-4”	18.5”	?	?	?	?	.199	.710	-	?	?	Heating coil
T	Nozzle	2”	252’-8”	18”	?	?	?	?	.200	.655	-	?	?	Heating coil
U	Nozzle	2”	253’-10”	18.5”	?	?	?	?	.191	.715	-	?	?	Heating coil
V	Nozzle	2”	255’-2”	18.5”	?	?	?	?	.217	.720	-	?	?	Heating coil
W	Couplet	1.25”	261’-9”	65”	?	?	?	?	-	-	-	?	?	Temp. probe (not used)



*** - Riveted**

SHELL NOZZLE AND APPURTENANCES TABLE (Cont.)

Item	Description	Size	Location	CL Elev.	Repad Shape	Repad Height	Repad Width	Repad Thk.	Neck Thk.	Flange Thk.	Cover Thk.	Weld Space	Tell Tale	Comments
X	Nozzle	16”	274’-0”	25”	?	?	?	?	.373	1.478	-	?	?	Inlet w/ diffuser
Y	Couplet	2”	281’-3”	6”	?	?	?	?	-	-	-	?	?	Plugged
Z	Butt Insert	8”	282’-1”	8”	?	?	?	?	.505	-	-	?	?	
AA	Nozzle	2”	317’-0”	14.5”	?	?	?	?	.145	.813	-	?	?	Waterdraw
BB	Nozzle	2”	320’-4”	7”	?	?	?	?	.189	.672	-	?	?	Heating coil
CC	Nozzle	2”	321’-8”	7”	?	?	?	?	.230	.720	-	?	?	Heating coil
DD	Nozzle	2”	322’-10”	7”	?	?	?	?	.226	.717	-	?	?	Heating coil
EE	Nozzle	2”	324’-3”	7”	?	?	?	?	.217	.656	-	?	?	Heating coil
FF	Nozzle	2”	325’-6”	7”	?	?	?	?	.224	.670	-	?	?	Heating coil



*** - Riveted**

FIXED ROOF INSPECTION

Item	Description	Comments
1	Inspect fixed roof protective coating system.	• N/A – No exam, due to replacement.
2	Inspect frangible joint / eave angle for corrosion, defects and coating failure.	• The eave angle (L4" / Lap welded / Inward turned) has a corrosion hole, in the top of the angle. The hole is located above manway "A". Recommend repairing when the roof is replaced.
3	Inspect fixed roof for low spots and external corrosion.	• N/A – No exam, due to replacement.
4	Conduct visual / UT inspection on roof nozzles and appurtenances.	• No exam of the roof plates or nozzles. The roof plates have severe underside thinning with hundreds of through plate holes. The thinning is through out the entire roof. Due to the condition of the roof, the decking is to be replaced at this out-of-service. Note: The floor has significant ponding after each rain.
5	Visual inspection of internal roof rafters, girders, columns and bases.	• A total of twenty (20) rafters are bent / twisted. The roof has a total of one-hundred and forty-four (144) rafters. Location of distorted rafters: (5) center section, (3) middle section and (12) outer section. Recommend replacing these rafters at a minimum. Also, recommend performing a visual exam on the rafters for thinning, when the roof decking is replaced. • No. Rafters Row 1: 24 (From Center) • No. Rafters Row 2: 48 • No. Rafters Row 3: 72
6	Verify open roof vents have screens and covers.	• N/A – No open top roof vents.
7	Check condition of vents, missing or clogged screens. Note possible insufficient venting	• N/A – No exam, due to replacement.
8	Verify handrails and tie-off supports are in good condition.	• N/A – No exam, due to replacement.
9	Measure any significant bulges and creases in the roof plates.	• N/A – No exam, due to replacement.
10	Do any roof nozzles or appurtenances have bolt-on type reinforcement pads?	• N/A – No exam, due to replacement.
11	Has the roof been converted from an external to internal with a Geodesic Dome?	• N/A – Lap welded fixed roof.

FIXED ROOF INSPECTION (Cont.)

Item	Description	Comments
12	Conduct visual inspection of roof insulation.	• N/A – No roof insulation. The roof is painted black.

Additional Notes:

- Note: The rafter and shell clip (L4" angle) above floor plate #110 was inspected by using a manlift. The rafter had no visible corrosion or thinning and the support clip was in good condition. Also, the rafters on each side appeared to be in good structural condition. However, recommend visually inspecting all rafters, girders and supports when the roof decking is replaced. It is believed the rafters could be thinning, in the areas of the through plate holes.

SECTION 3

NON-DESTRUCTIVE TESTING (NDT)

NDT INSPECTION SCOPE-OF-WORK

The following Non-Destructive Testing examinations were performed to evaluate the integrity of the tank.

- Magnetic Flux Leakage (MFL) floorscan of the tank bottom, for significant internal and underside metal loss.
- A Visual Test (VT) was performed in areas inaccessible to the MFL, floor plate lap rivets, interior corner angle (riveted), under low internal piping and areas of concern.
- Ultrasonic Testing (UT) of MFL indications, for remaining plate thickness.
- Ultrasonic (UT) thickness readings of the tank shell courses
- Ultrasonic (UT) thickness readings of the shell nozzles.
- Internal settlement survey

NDT SIGNIFICANT FINDINGS

The MFL floorscan and visual examination revealed a total of three-hundred & thirty six (336) recordable areas. A breakdown of these areas is as follows:

(03) Underside Corrosion
(333) Topside Pitting

RECORDING CRITERIA

Underside areas of metal loss equal to or less than 0.150" remaining plate thickness as established by the client were considered recordable. Topside metal loss areas equal to or less than 0.120" remaining plate thickness were recorded for repairs.

Recordable areas of underside metal loss were marked with a pink square paint border and white permanent marker. Refer to the "Bottom Plate Reduction Table" for data.

Recordable areas of internal metal loss were marked with a pink round paint border and white permanent marker. Refer to the "Bottom Plate Reduction Table" for data.

FLOOR EXAMINATION

The floor plates were examined utilizing the MFL floorscan, Ultrasonic Testing (UT) and Visual Testing (VT) methods. Areas identified by the MFL as exceeding or closely approaching the critical thickness was quantified by Ultrasonics and pit gauge measurements.

Limitations to the MFL floorscan exam included: near the floor-to-shell rivet angle, floor plate lap rivets, under fixed roof supports, sumps, puddle welds and under low internal piping.

FLOOR PLATE LAP RIVET EXAMINATION

The floor plate lap rivets were inspected utilizing visual testing methods. This exam revealed no recordable leakage.

FLOOR-TO-SHELL ANGLE (riveted) EXAMINATION

The internal floor-to-shell angle was inspected utilizing visual testing methods. The exam revealed no recordable leakage or thinning.

SHELL COURSE ULTRASONIC THICKNESS READINGS

Ultrasonic thickness readings were taken along one (1) vertical drop location, with three (03) thickness readings per course. Refer to the "Shell Plate Ultrasonic Thickness Readings" for data.

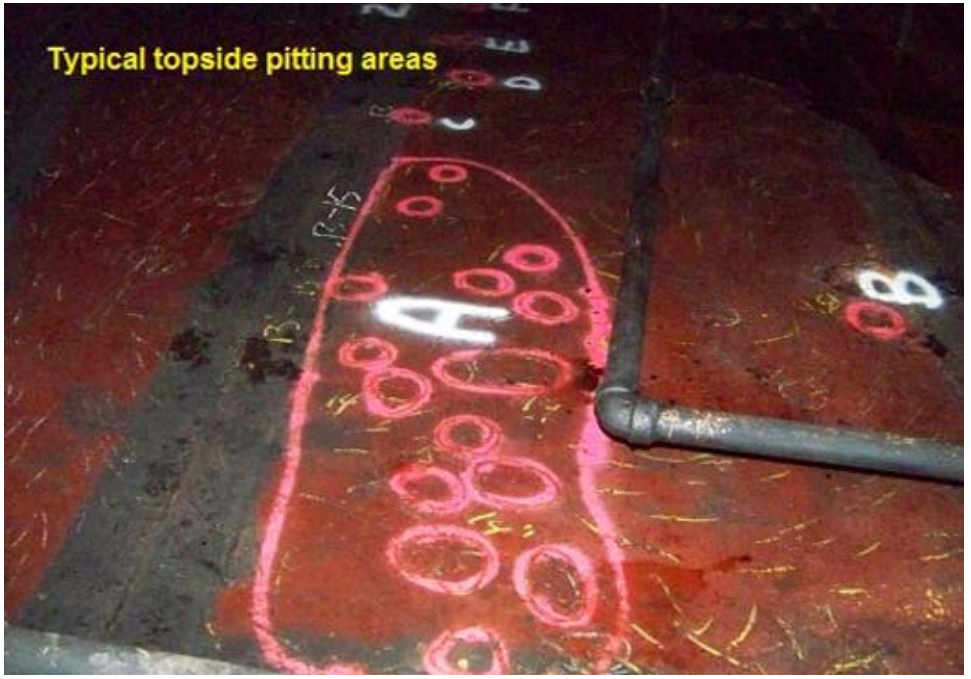
SHELL NOZZLE EXAMINATION

The shell nozzles, reinforcing pads, covers and flanges were inspected by Ultrasonic and Visual Testing methods. Refer to the "Shell Nozzles and Appurtenances" for data**.**

SECTION 4

Pictures


A
Typical underside corrosion area


Typical topside pitting areas
A

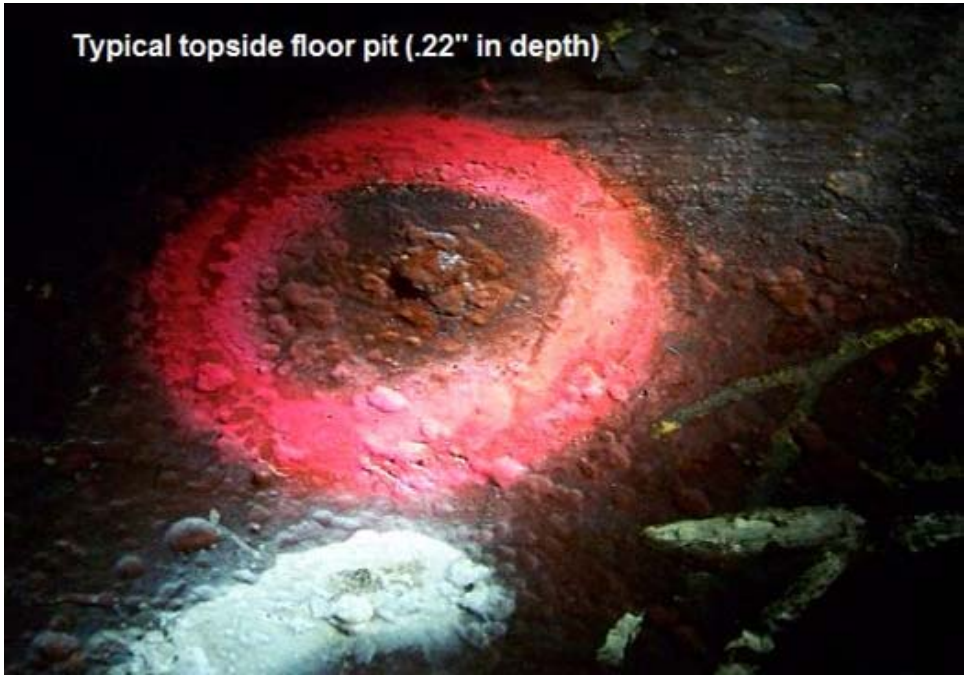

Distortions / damaged insulation

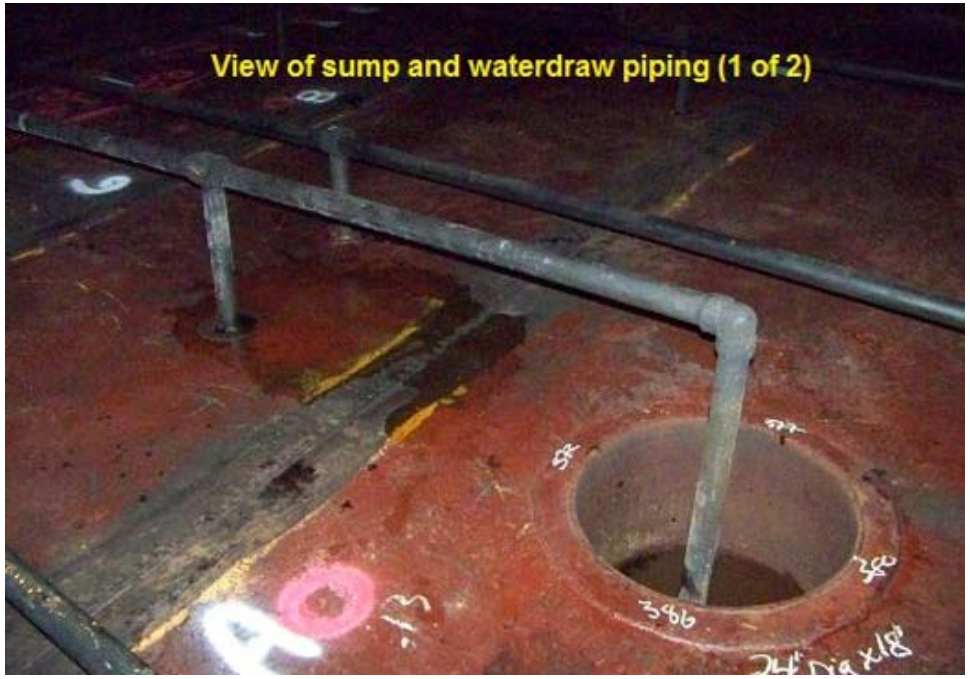

Typical underside corrosion area
A


Topside pitting areas

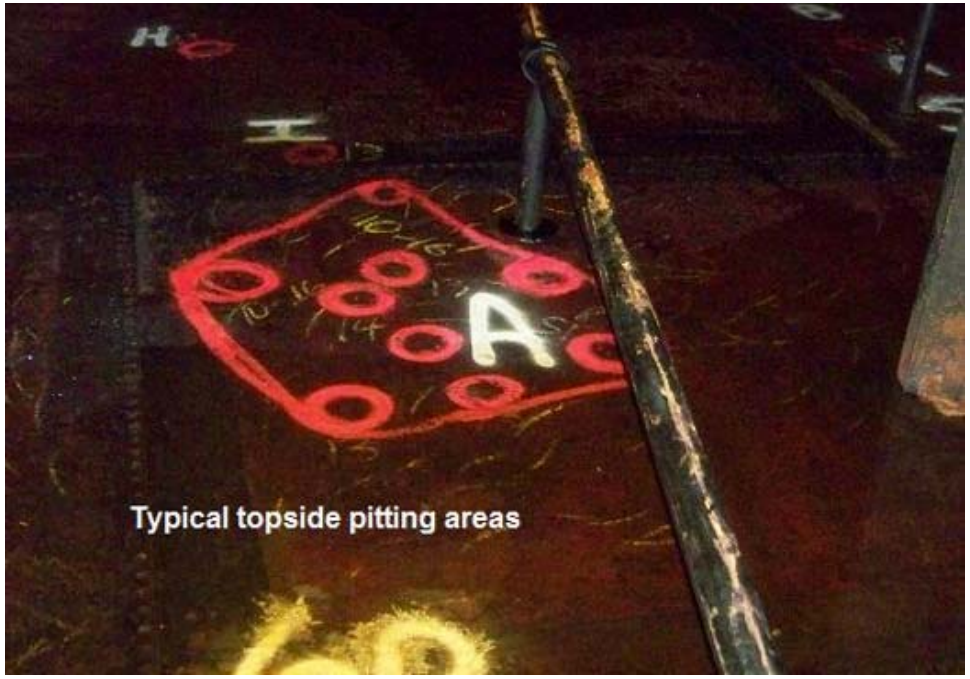

Typical floor pitting through out the tank


Topside pitting areas


Typical topside floor pit (.22" in depth)


View of sump and waterdraw piping (1 of 2)

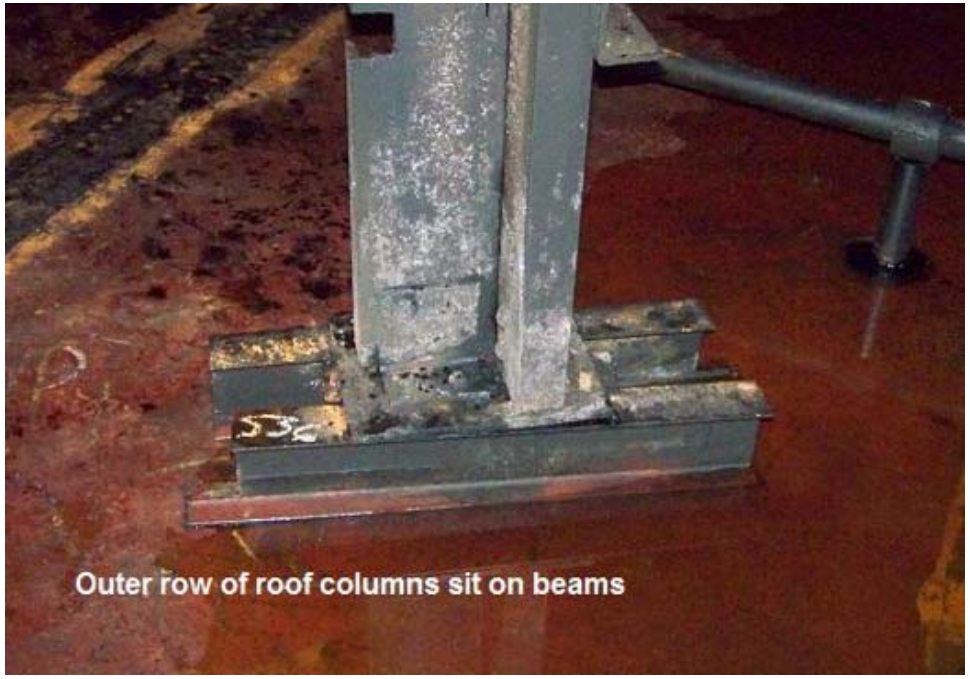

View of fixed roof support column / No bearing pad
or lateral retaining clips


Typical topside pitting areas


Heating coil support pads not seal welded (165 total)

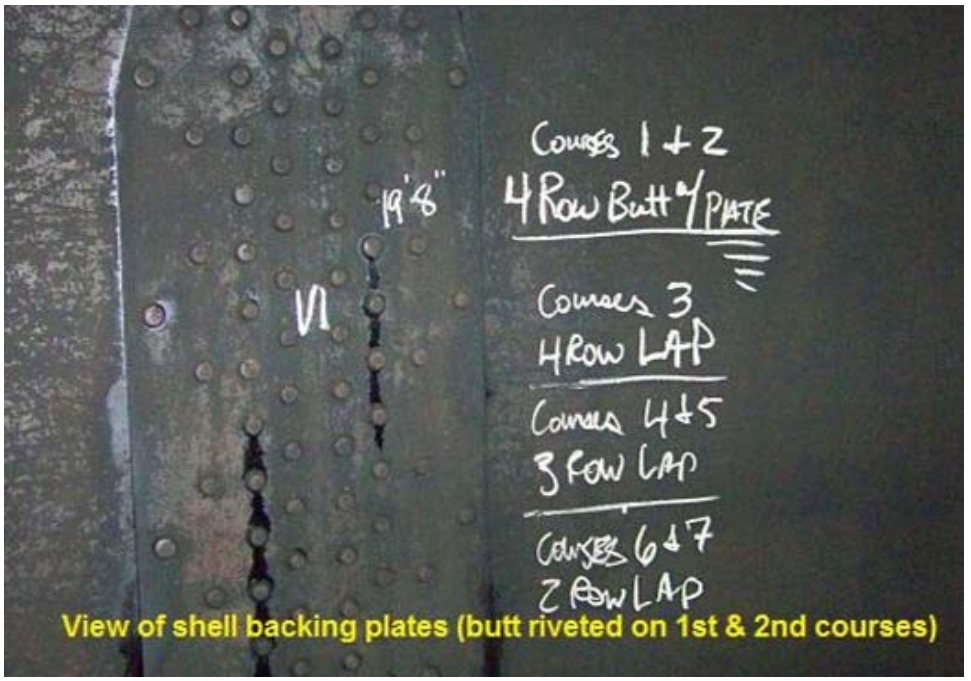

Standing water on floor


Through plate holes in roof plates


Outer row of roof columns sit on beams

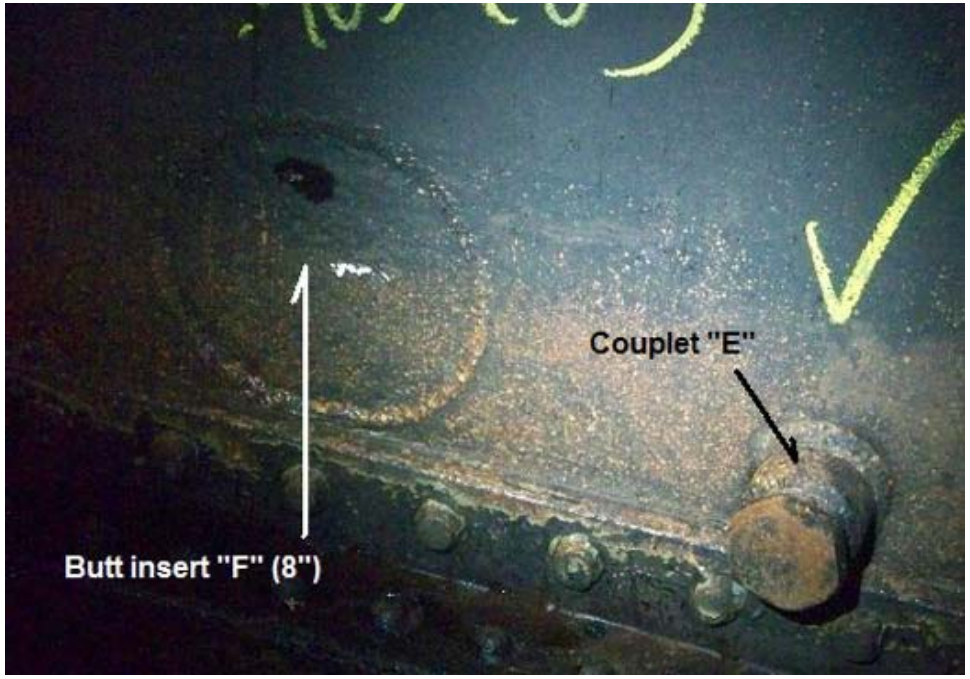

Standing water on floor

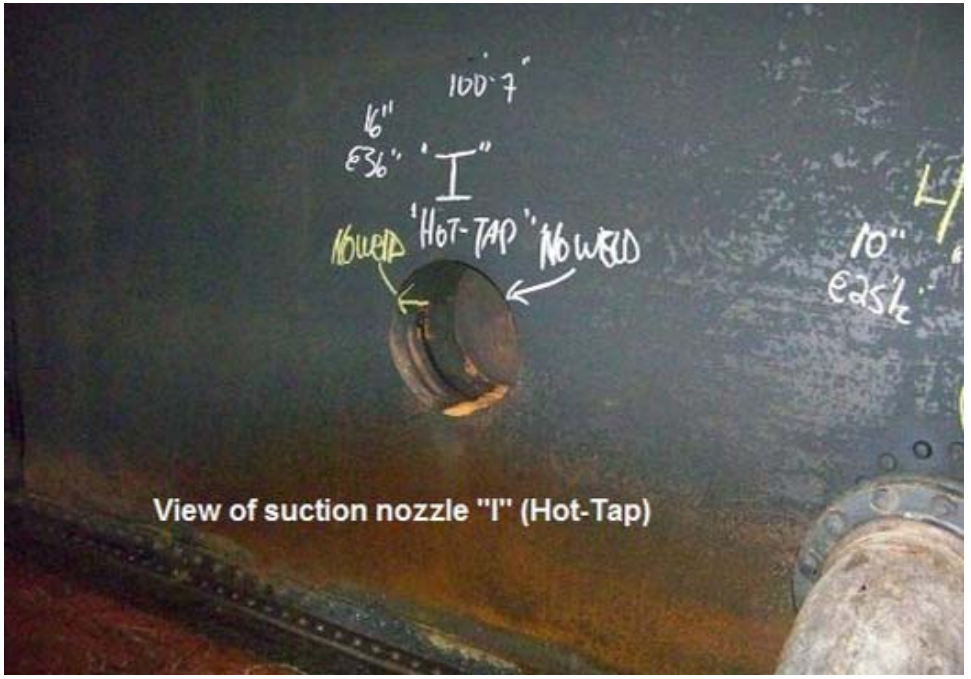

View of floor-to-shell angle (L4")

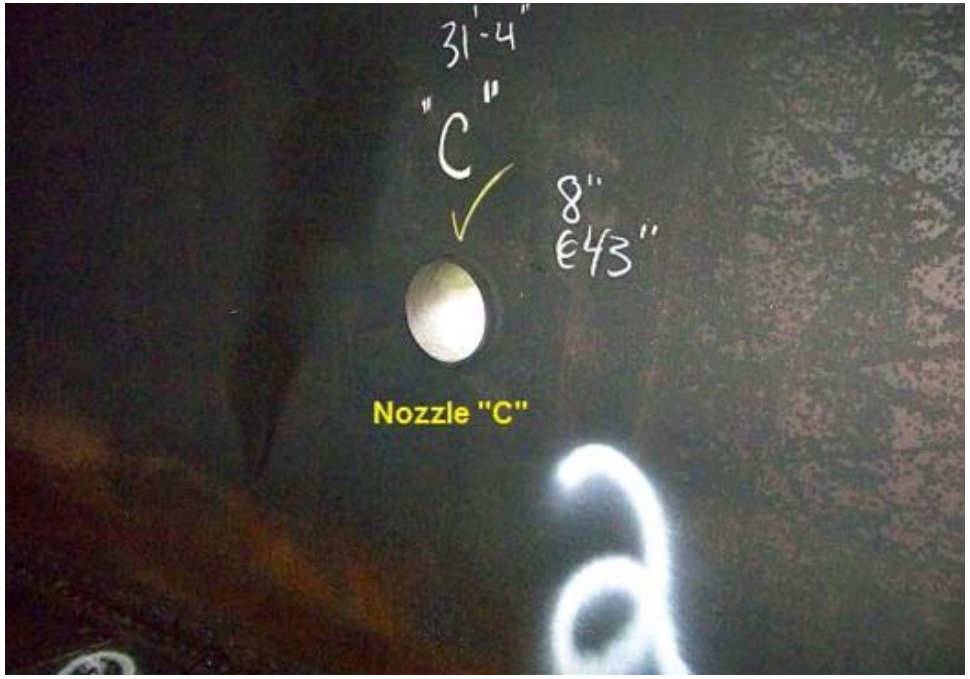

Courses 1 + 2
4 Row Butt w/ PLATE
19'8"
VI
Courses 3
4 Row LAP
Courses 4 + 5
3 Row LAP
Courses 6 + 7
2 Row LAP
View of shell backing plates (butt riveted on 1st & 2nd courses)


Through plate holes in roof

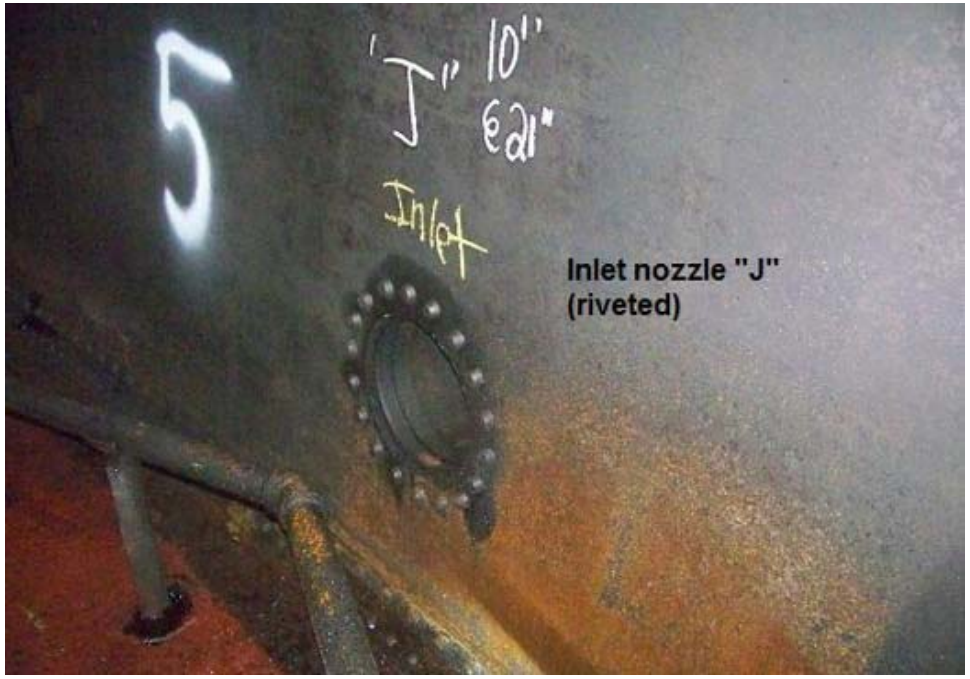

Epoxy coating at rivet seams

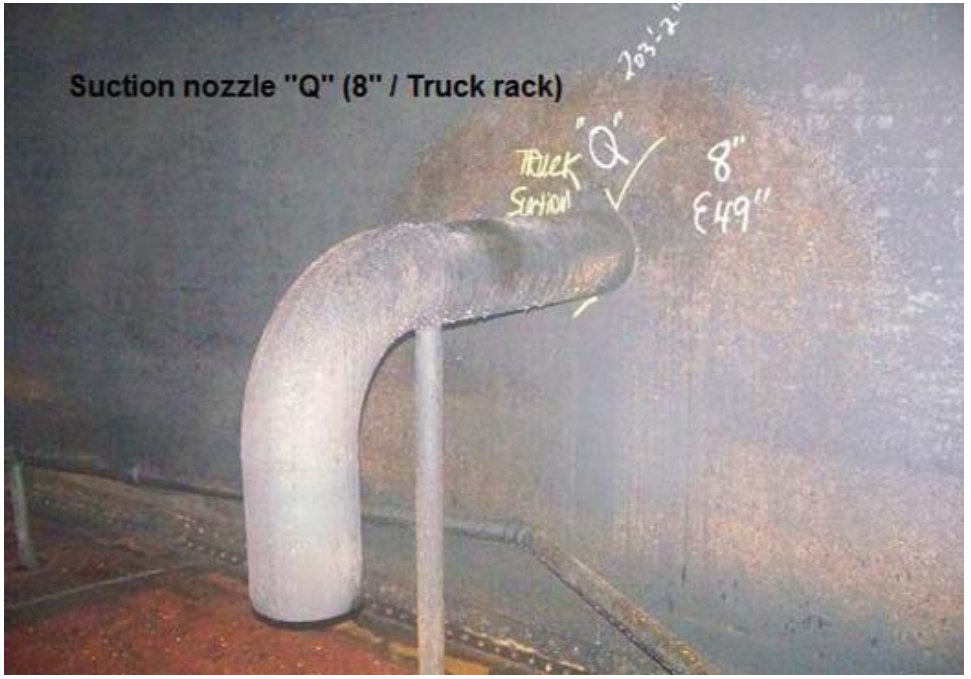

Couplet "E"
Butt insert "F" (8")

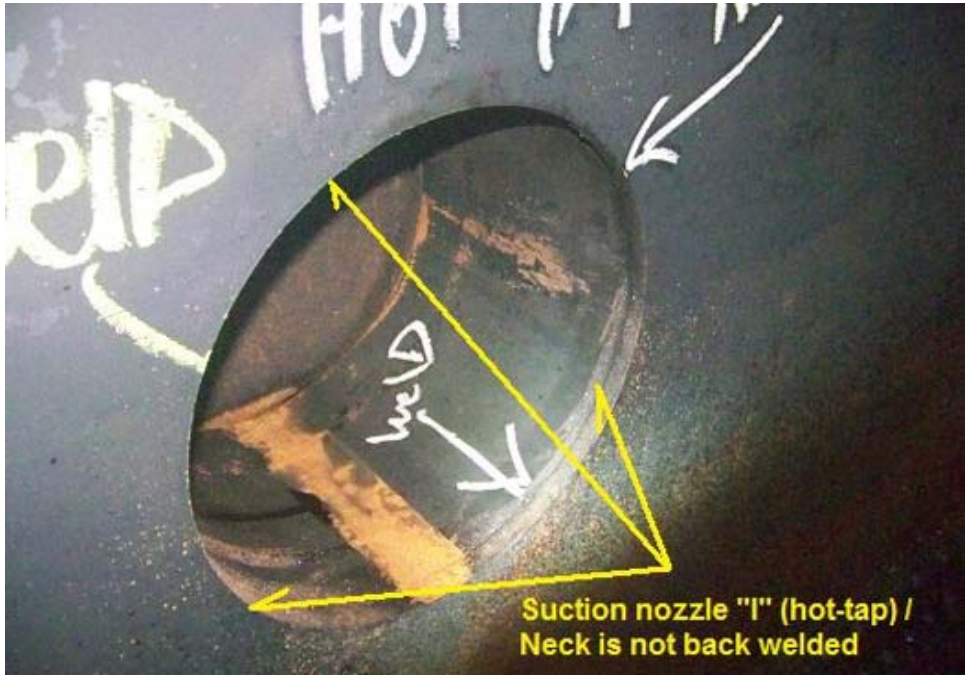

100'-7"
16"
E36"
"I"
"HOT-TAP" NO WELD
10"
E25½"
View of suction nozzle "I" (Hot-Tap)

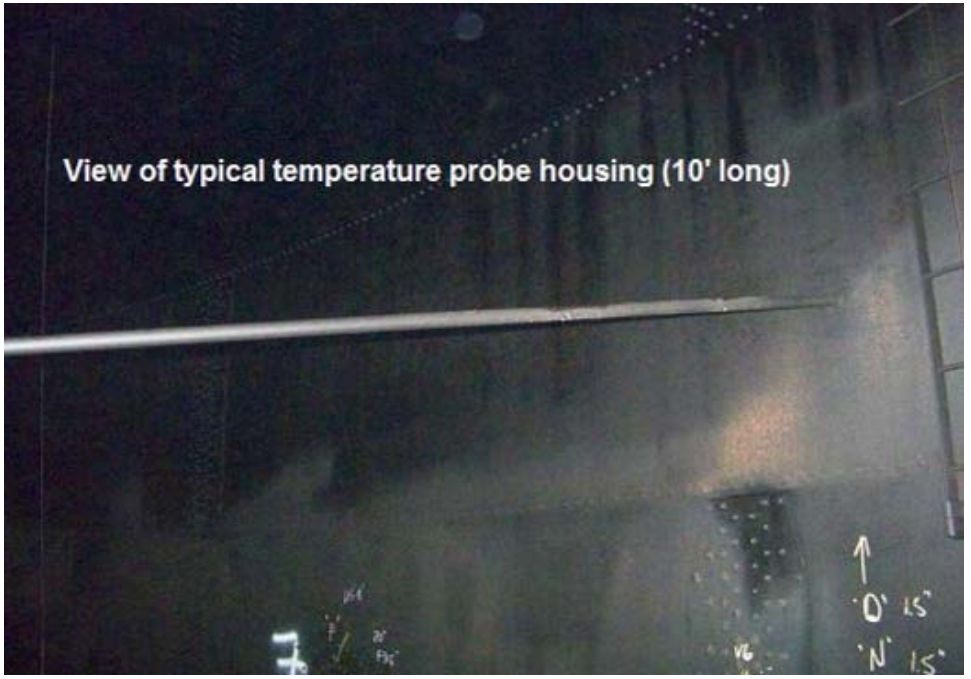

31'-4"
"C"
8"
E43"
Nozzle "C"

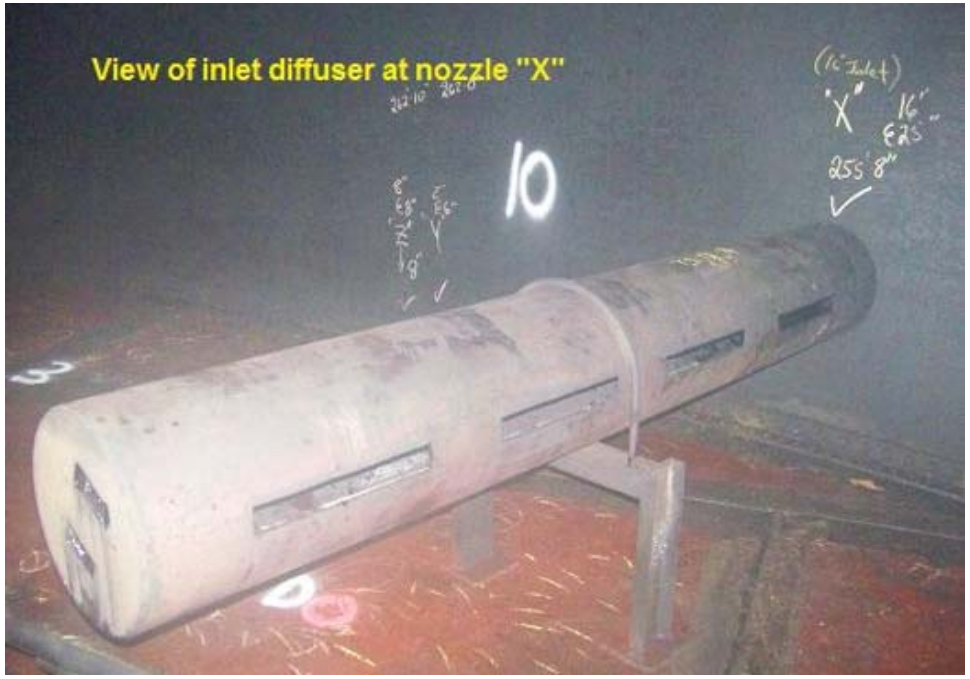

E25½"
H
View of low suction piping

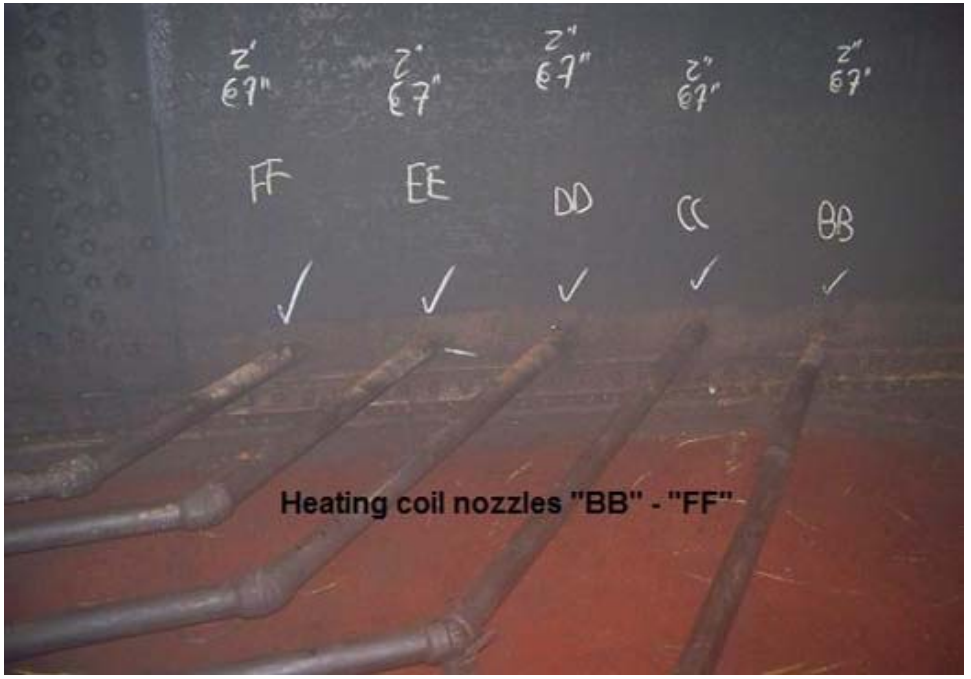

5
"J" 10"
E21"
Inlet
Inlet nozzle "J"
(riveted)

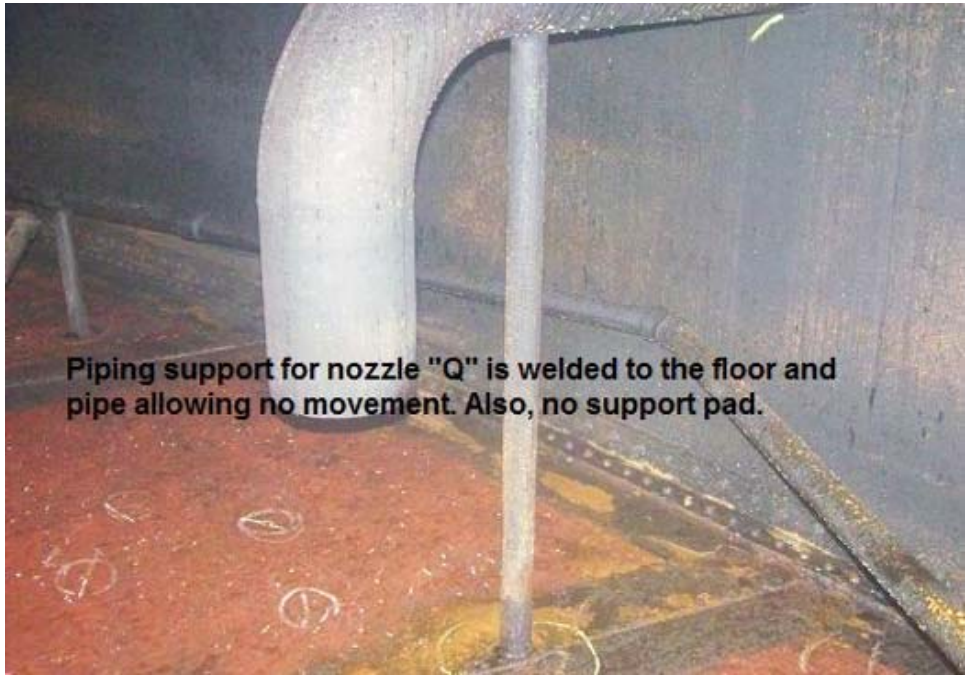

Suction nozzle "Q" (8" / Truck rack)
Truck Suction
"Q"
8"
E49"

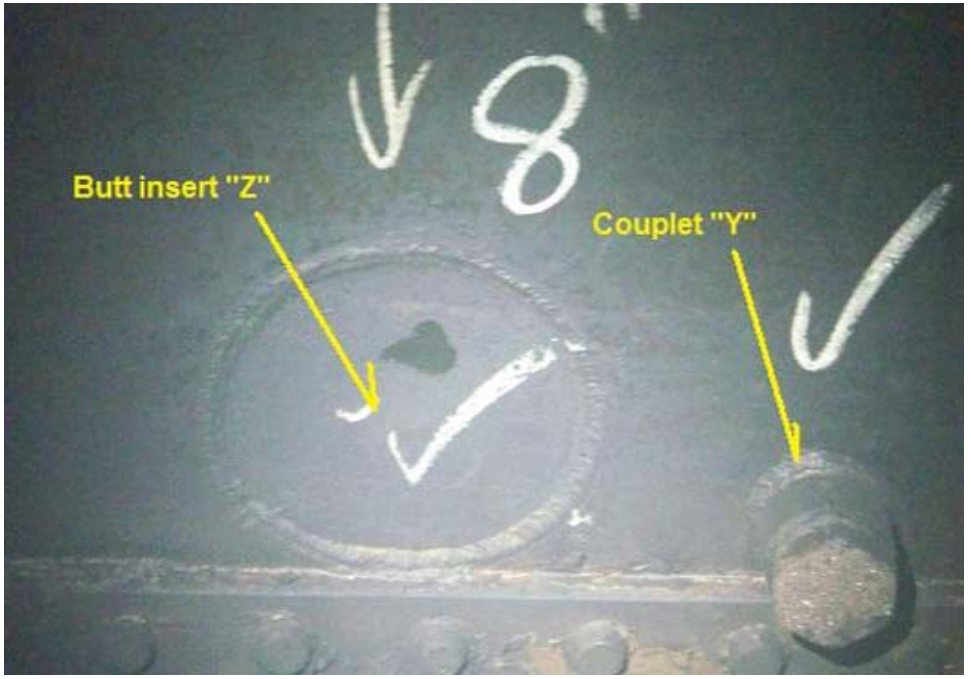

weld
Suction nozzle "I" (hot-tap) /
Neck is not back welded


View of typical temperature probe housing (10' long)
"O" LS
"N" LS


View of inlet diffuser at nozzle "X"

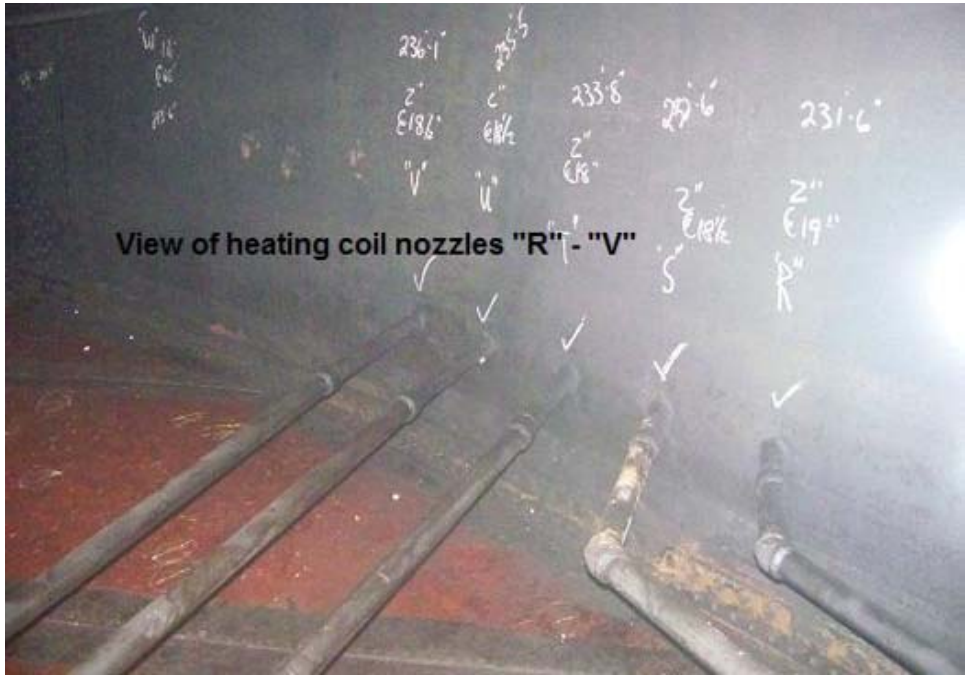

FF
EE
DD
CC
BB
Heating coil nozzles "BB" - "FF"


Piping support for nozzle "Q" is welded to the floor and pipe allowing no movement. Also, no support pad.


Butt insert "Z"
Couplet "Y"


Overall View / Tank #201


Manway "A"


View of heating coil nozzles "R" - "V"


View of mixer (Manway "A")


C
Nozzle "C" (8" / Not in use)

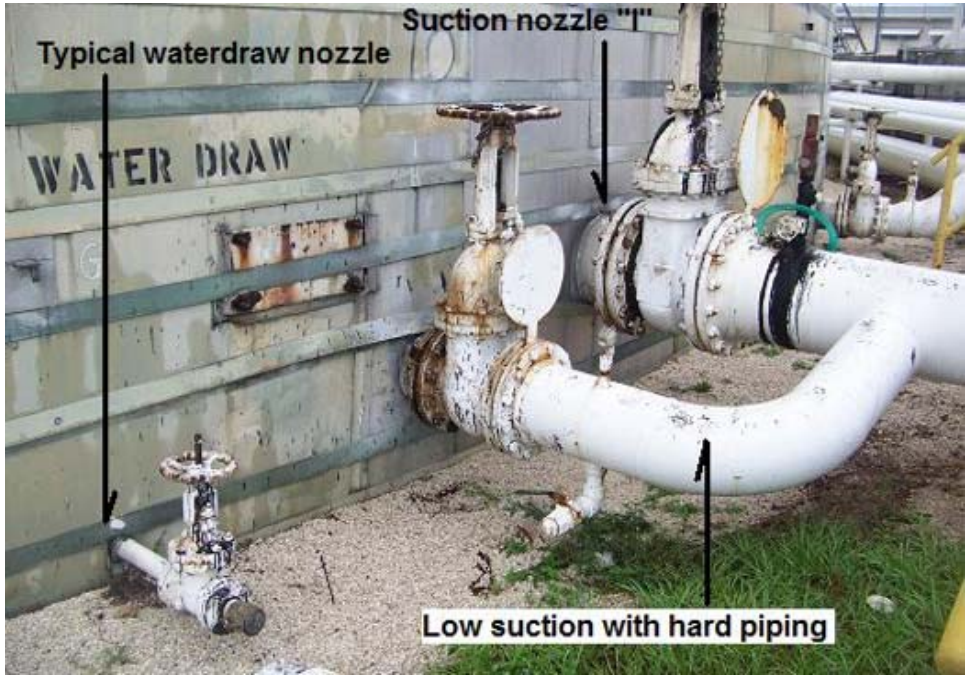

View of earth foundation / Tank sits below the grade


Product stains on shell

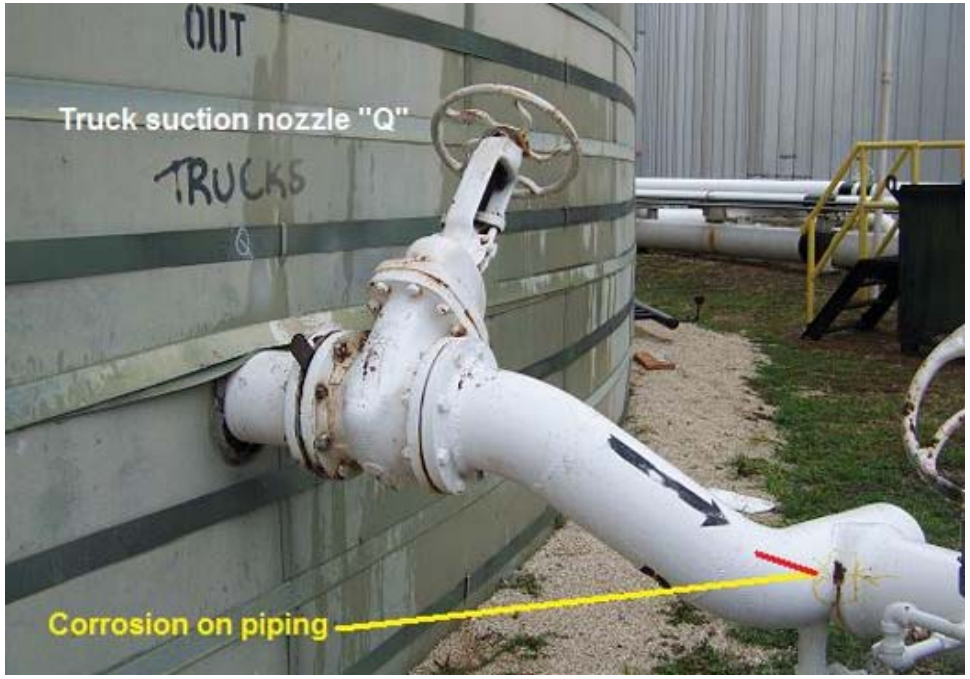

Coating failure and minor corrosion on shell


Product stain on pressure relief at suction nozzle


View of radial stairway


Typical waterdraw nozzle
Suction nozzle "I"
WATER DRAW
Low suction with hard piping

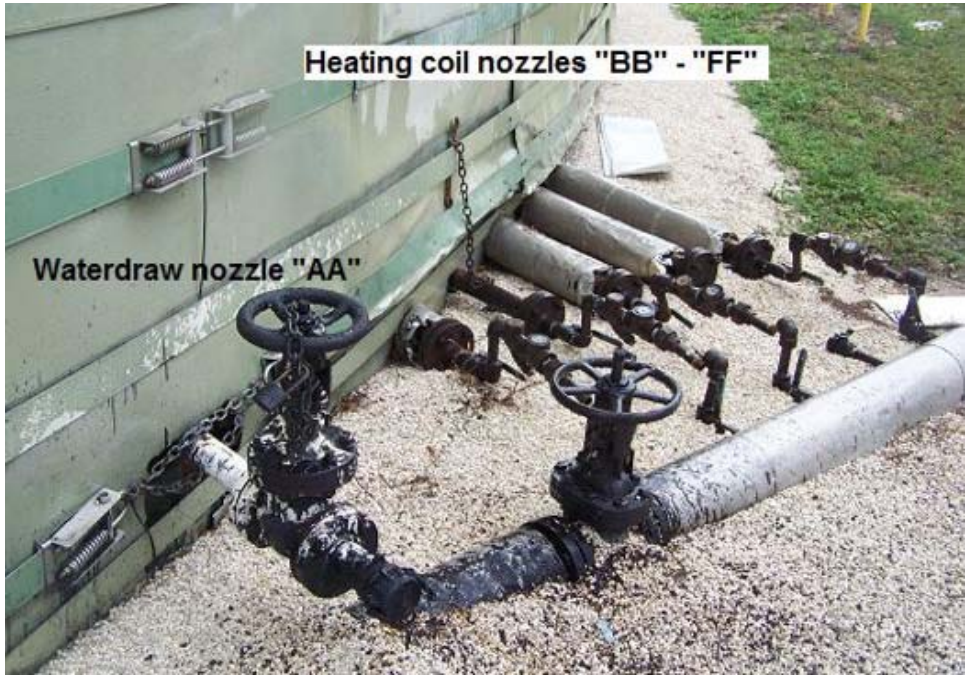

Inlet nozzle "J"


OUT
Truck suction nozzle "Q"
TRUCKS
Corrosion on piping


Heating coil nozzles "R" - "V"

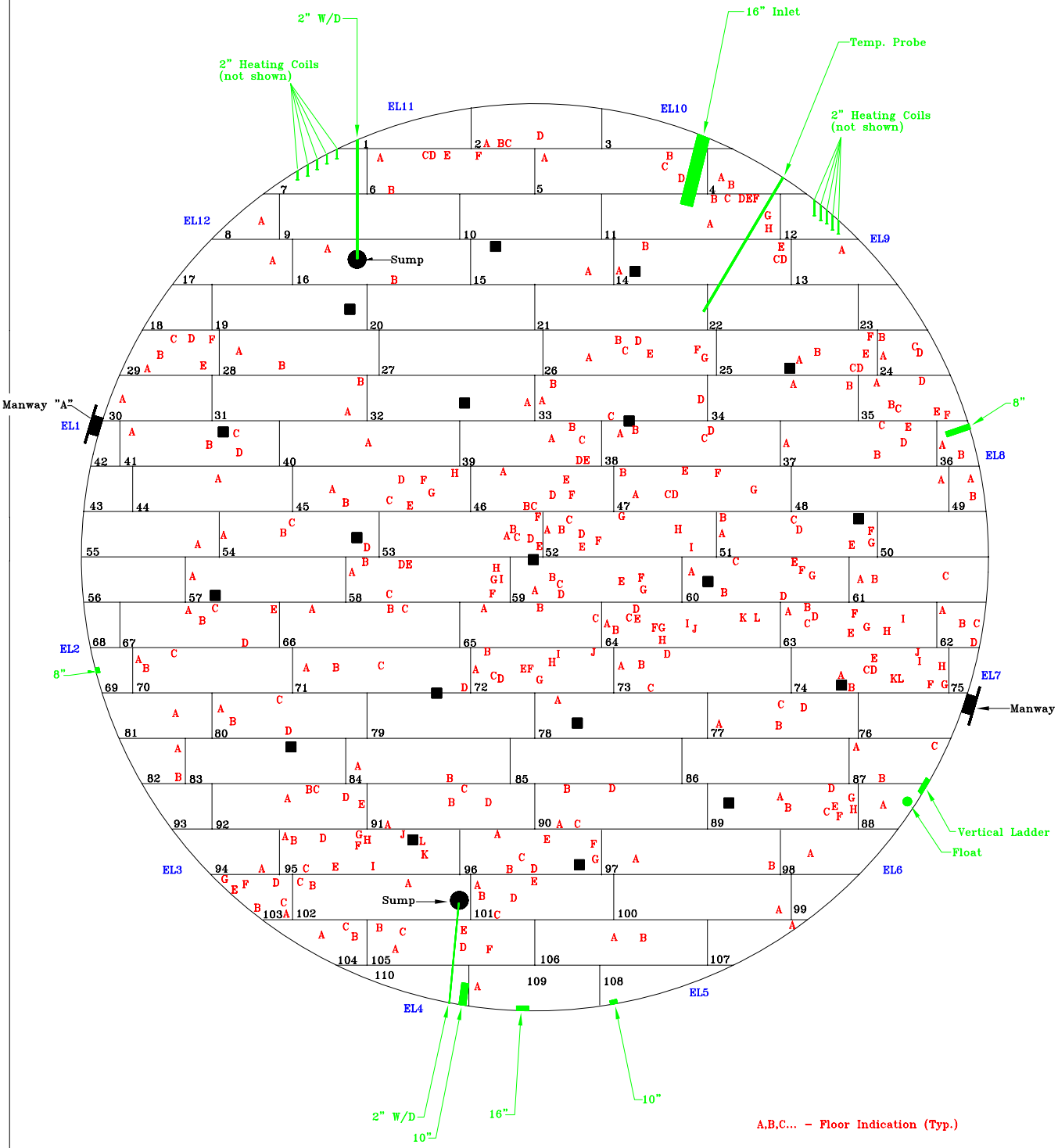

Varec gauge
Vertical Ladder
#6 OIL


Corrosion on nozzle "Q" piping (Truck suction)


Heating coil nozzles "BB" - "FF"
Waterdraw nozzle "AA"


X
16
Inlet nozzle "X"


Distortions in insulation



A,B,C... – Floor Indication (Typ.)

■ – Fixed Roof Column (Typ.)

EL# – Elevation Location (Typ.)

Note: Heating coils not shown

SUMP THICKNESS READINGS

PLATE #16

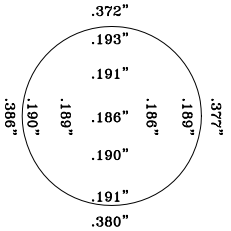
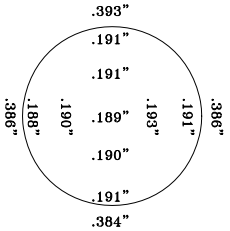


PLATE #102



SEMPER FI ENTERPRISES LLC

INSPECTION DIVISION

Tank Number: 201

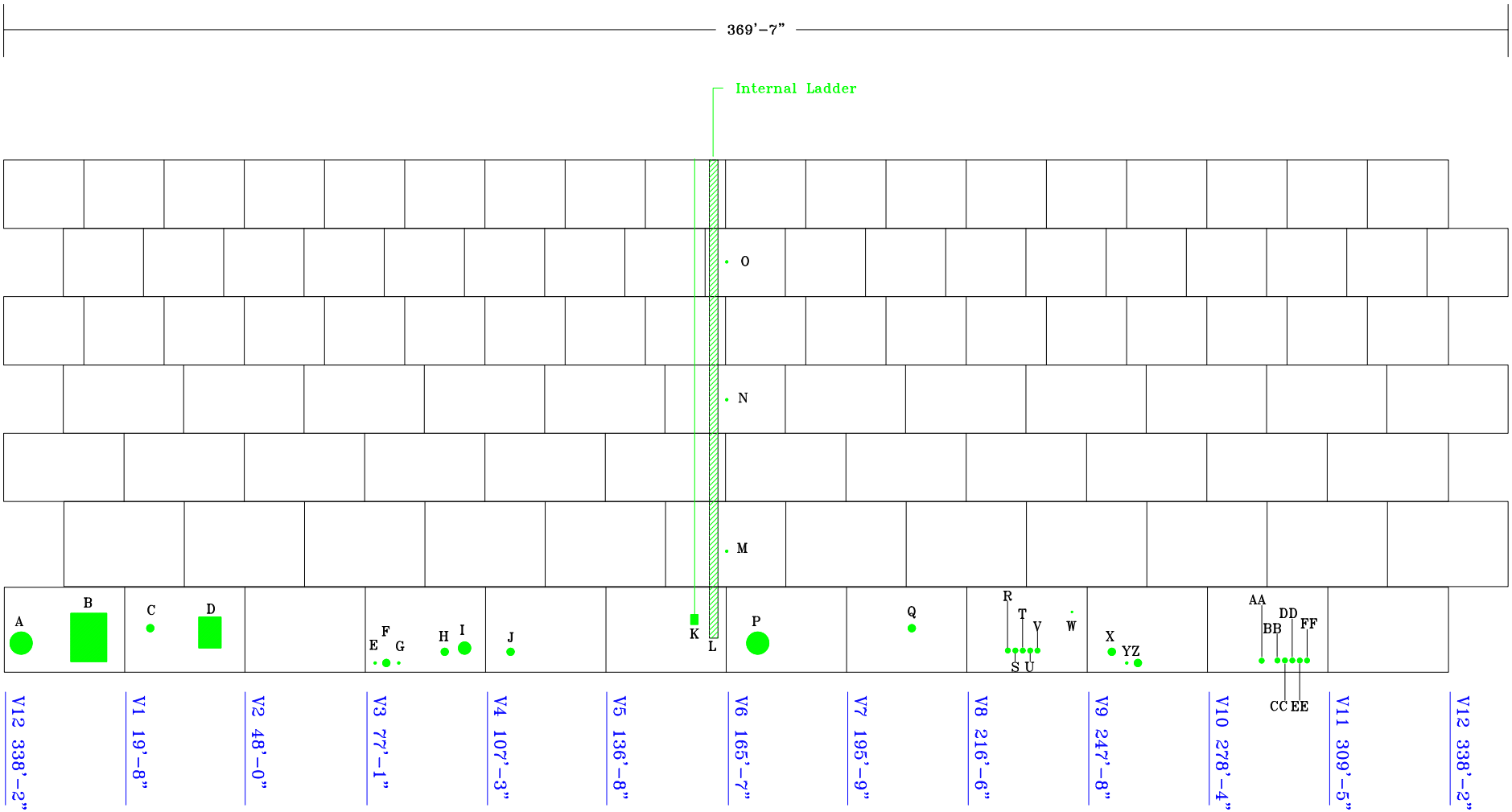
Drawing: FLOOR PLATE LAYOUT

Owner: TRANSMONTAIGNE

Location: PORT EVERGLADES, FL

DWG. NO.: 5–1

DATE: 06–23–11



NOTE: Shell was drawn from inside the tank

SEMPER FI ENTERPRISES LLC

Inspection Division

OWNER: TRANSMONTAIGNE	TANK NO.: 201
DRAWING : SHELL PLATE LAYOUT	DIAMETER: 117'-0"
LOCATION : PORT EVERGLADES, FL	HEIGHT: 42'-0"
DRAWING NO. : 5-2	DATE: 06-25-2011